



Faculty Profile on University Website

www.uprtou.ac.in

Title	Dr.	First Name	DINESH	Last Name	GUPTA	Photograph
Designation	ASSISTANT PROFESSOR- CHEMISTRY (Contract)					
School	SCHOOL OF SCIENCE					
Address (Campus)	SCHOOL OF SCIENCE, U.P. RAJARSHI TANDON OPEN UNIVERSITY, PRAYAGRAJ, U.P. , INDIA-211021					
(Residence)	B-42 GOVINDPUR COLONY, PRAYAGRAJ, INDIA-211004					
Phone No (Campus)	+91 7525048053					
(Residence)	+91 9450622596					
Mobile	+91 9450622596, +91 8318920132					
Fax						
Email	guptadines[at]gmail.com, dkgupta[at]uprtou.ac.in					
Web-Page	http://orcid.org/0000-0001-8724-6326 , https://www.researchgate.net/profile/Dinesh_Gupta2?ev=hdr_xprf https://scholar.google.co.in/citations?user=8x5ld0AAAAJ&hl=en http://www.uprtou.ac.in/cv/science/dkgupta.pdf Scopus Author ID: 57191032946					<u>Citations</u> 136 <u>h-index</u> 4 <u>i10-index</u> 1
Educational Qualifications (Graduation Onwards)						
Course/Degree	Institution			Year	Details/Thesis Topic/Subjects	
B.Sc.	University of Allahabad			2003	Botany and Chemistry	
M.Sc.	University of Allahabad			2005	Inorganic Chemistry	
Ph.D.	Motilal Nehru National Institute of Technology, Allahabad			2012	Synthesis and Characterization of some Modified Metal Alkoxides as Precursors for Metal Oxide Nanomaterials and some Metal Organic Framework Compounds	
UGC/CSIR-NET/JRF/SRF						
Roll No	Subject				Year	
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Career Profile						
Organization / Institution			Designation	Duration	Role	
Bharat Scout & Guide Inter College, Allahabad, India			Lecturer- Chemistry	2005-2006	Teaching and Administrative work	
Chhatrapati Shahuji Maharaj College of Engg. and Technology, Iradatganj Allahabad, India			Lecturer- Chemistry	2008-2012	Teaching and Administrative work	

U.P. Rajarshi Tandon Open University, Prayagraj, India	ASSISTANT PROFESSOR- CHEMISTRY (Contract)	Feb. 2013 to Till date	Teaching and Administrative work
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Research Interests / Specialization

1. Designing and synthesizing various Metal Organic Framework complex formation using various transition metal ions.
2. Designing and synthesizing various Metal Alkoxides and study properties of their sol-gel synthesized nanomaterials.
3. Characterization of ligands and metal complexes using Single crystal X- ray crystallography, powder X-ray of Nanomaterials and various spectroscopic techniques like multinuclear NMRs and EPR, Titration techniques, SEM, TEM and Fluorescence.
4. Instruments like FT-IR, UV-Vis spectrophotometer, Elemental analyzer, TGA- DSC, Partical size analyzer (Nano Track), Dip Coater and Spin Coater, Schlenk Technique, Globe Box Experiments etc.

Contribution to design and development of Self Learning Material (SLM) of the programmes

Programme Name	Course Code	Course title	Curriculum designing (Programme and course coordination)	Writing of No. of Units	Editing of blocks (content format and language)	Vettin g of blocks
B.Sc.	UGCHE-101	Basic Inorganic Chemistry (Inorganic Chemistry-1)	B.Sc. -Chemistry, Dr. Dinesh Kumar Gupta	6	--	--
B.Sc.	UGCHE-103	Basic Physical Chemistry (Physical Chemistry-1)	B.Sc. -Chemistry, Dr. Dinesh Kumar Gupta	3	--	--
B.Sc.	UGCHE-104	Inorganic Chemistry II (Advance Inorganic Chemistry)	B.Sc. -Chemistry, Dr. Dinesh Kumar Gupta	3		
B.Sc.	UGCHE-07	Inorganic Chemistry III (Selected Topics In Inorganic Chemistry)	B.Sc. -Chemistry, Dr. Dinesh Kumar Gupta	1	--	--
B.Sc.	UGCHE-10	Advanced Analytical Techniques	B.Sc. -Chemistry, Dr. Dinesh Kumar Gupta	1	--	--
B.Sc.	Solid Waste Management (SWM)	Sources and Types of Municipal Solid Waste	B.Sc. -Chemistry, Dr. Dinesh Kumar Gupta	3	--	--

Honors & Awards

Year of the Award/appointment	Name of the award or fellowship	Incentive given by OU in recognition of the award	Link for the relevant documents
2021	Sikshak Samman, By U. P State Government, U.P. India		

Publications /Academic Activities (Numbers Only)

Books & Monographs (Single Author)	2	Research Papers Published in International Journals	4	Papers Presented in Seminars/ Conferences	12	Seminars/ Conferences Organized	2	Research Projects (Comple ted)	
Books (Co-authored)		Research Papers Published in	1	Seminar/ Conferences Attended		Workshops Organized		Research Projects (Ongoing)	

		Other Journals							
Books (Edited)	1	Articles Published in Popular Fora, e.g., Websites, Blogs, Newspapers, Magazines etc.		Sessions Chaired in Seminars/ Conferences	1	Membership of Academic/ Professional Bodies	4	Foreign Countries Visited for Academic Assignments	
Chapters in Edited Books	4			Resource Lectures Delivered					
Publication on Distance Education									

Details of Publications /Academic Activities						
a) Books / Monographs/Book chapters						
S.N	Year	Name of the Authors/ Editors	Title of the Book/ Edited Book	ISBN number	Name of the Publisher	Co- Author
	2022	Dr. Dinesh Kumar Gupta	DCECHE -106: INORGANIC CHEMISTRY III (SELECTED TOPICS IN INORGANIC CHEMISTRY) Dr. Dinesh Kumar Gupta , Author (Block-1 Unit-2; Block-2 Unit- 7 and 8) Academic Consultant- Chemistry School of Science, UPRTOU, Prayagraj, U. P., India	978-93-94487-33-8	UPRTOU, Prayagraj.	
	2022	Dr. Dinesh Kumar Gupta	SBSCHE-01 ORGANIC CHEMISTRY II (ADVANCE ORGANIC CHEMISTRY) Author Block-1 Unit-1,2 and 3) Dr. Dinesh Kumar Gupta Academic Consultant- Chemistry, School of Science, UPRTOU, Prayagraj, U. P. India	978-81-953893-0-8	UPRTOU, Prayagraj.	
	2021	Santosh Kumar, Dinesh Kumar Gupta, Joonseok Koh, Rohit Srivastava	Book Title- Nano-catalysts for Energy Applications Chapter-3 Porphyrins Based Nanostructured Material for the Conversion of CO ₂ into Value Added Products	ISBN: 978-0-367-53643-5	CRC Press of Taylor & Francis Group, LLC	Santosh Kumar, Joonseok Koh, Rohit Srivastava
	2021	Dr. Dinesh Kumar Gupta	UGCHE-104 :Inorganic Chemistry II (Advance Inorganic Chemistry) Block-02	Under Allotment	UPRTOU, Prayagraj.	

			Unit- 4 Chemistry of Lanthanides and Actinides Unit- 5 Chemistry of Nobel Gases Unit- 6 Acid - Base and Non-aqueous solvents			
	2020	Dr. Dinesh Kumar Gupta	UGCHE-103 : Basic Physical Chemistry (Physical Chemistry-1) Block-01 Unit- 1 Mathematical Concepts and Computers Unit- 2 Gaseous and Liquid States Unit- 3 Solid State	Under Allotment	UPRTOU, Prayagraj.	
	2020	Dr. D. K. Gupta, Dr. Shruti, Prof. Ashutosh Gupta and Prof. G. S. Shukla (Editors)	Book Title: Emerging Trends of Biological Sciences and Bio-Statistics (Edited Book)	ISBN: 978-99-94604-39-3	Edwin Group of Publications, Kathmandu, Nepal	
	2020	Dinesh Kumar Gupta	Book Title: Emerging Trends of Biological Sciences and Bio-Statistics Chapter- Toward Efficient Drug Delivery and Bio-Imagine Through Metal Organic Frameworks	ISBN: 978-99-94604-39-3	Edwin Group of Publications, Kathmandu, Nepal	
	2020	Dr. Dinesh Kumar Gupta	UGCHE-101 Basic Inorganic Chemistry (Inorganic Chemistry-1)	ISBN: 978-93-83328-28-4	UPRTOU, Prayagraj.	
	2020	Dr. Dinesh Kumar Gupta	Solid Waste Management ,SWM Unit- 1 Introduction to Municipal Solid Waste Management Unit- 2 Composition of Solid Waste Unit- 3 Principle of Solid Waste Management	Under Allotment	UPRTOU, Prayagraj.	
	2019	Dr. D. K. Gupta, Dr. Shruti and Dr. P. P. Dubey (Editors)	“Dimensions of Nanotechnology and Bio-Statistics” (Edited Book)	ISBN: 978-620-0-07999-2	LAP LAMBERT Academic Publishing, Mauritius, RU	
	2019	Dinesh Kumar Gupta	Book Title: Dimensions of Nanotechnology and Bio-Statistics Chapter- Concept of Nanotechnology	ISBN: 978-620-0-07999-2	LAP LAMBERT Academic Publishing, Mauritius, RU	
	2019	Dinesh Kumar Gupta, Santosh Kumar &	Book Title: Dimensions of Nanotechnology and Bio-Statistics Chapter- Development of Organic-Inorganic	ISBN: 978-620-0-07999-2	LAP LAMBERT Academic Publishing, Mauritius, RU	Santosh Kumar and D. K. Sahu

		D. K. Sahu	Hybrids and Modified Alkoxides Precursors for Functionalized Nanomaterials in Support of Adsorption and Catalytic Applications: An Approach			
	2019	Shruti and D.K. Gupta	Book Title: Dimensions of Nanotechnology and Bio-Statistics Introduction/ Editorial	ISBN: 978-620-0-07999-2	LAP LAMBERT Academic Publishing, Mauritius, RU	

(b) Papers Published in Indexed/ Peer Reviewed Journals

<u>Year of Publication</u>	<u>Title</u>	<u>Journal</u>	<u>ISSN/ Indexing</u>	<u>Co-Author (s) (if any)</u>
2011	<i>Hydrothermal Synthesis, Structure and Photoluminescent Property of a novel Sodium- pyrazinyltetrazole framework compound</i>	Inorganic Chemistry Communications, 14 (2011) 1485–1488	1387-7003 UGC No. 21221	Sadhana Singh, Peter Mayer, Ashutosh Pandey
2012	<i>Synthesis of superparamagnetic bare Fe₃O₄ nanostructures and core/shell (Fe₃O₄/alginate) nanocomposites</i>	Carbohydrate Polymers, 89, 3 (2012) 821–829	0144-8617 UGC No. 5035	Manish Srivastava, Jay Singh, Madhu Yashpal, R. K. Mishra, Shipra Tripathi and Animesh K. Ojha
2012	<i>Solution Electrophoresis Technique in the Study of Biologically Important Fe(III)/ Ni(II) – Norvaline Binary Complex</i>	Inorganic Chemistry: An Indian Journal, 7, 3 (2012) 99-103	0974 - 746X	Praveen P. Singh, R. K. P Singh
2013	<i>Synthesis, Characterization and Photoluminescent Property of a Trinucleated Cadmium (II) Coordination Polymer Involving In Situ Ligand Reaction</i>	Journal of Chemical Crystallography 43, 2 (2013) 82-90	1074-1542 UGC No. 21824	Sadhana Singh, H. Nöth and Ashutosh Pandey
2013	<i>Synthesis and structural characterization of some titanium butoxides modified with chloroacetic acids</i>	Transition Metal Chemistry, 38 (2013) 835-841	0340-4285 UGC No. 34925	Subia Ambreen, Kiran Gupta, Sadhana Singh, Stephane Daniele, N. D. Pandey, Ashutosh Pandey
2024	<i>MOF magic: Zirconium-based frameworks in theranostic and bio-imaging applications</i>	Journal of Materials Chemistry B	https://doi.org/10.1039/D3TB02562D	Santosh Kumar and Mohammad Younus Wani

(c) Publication on Distance Education

<u>Title of Book/chapters published/edited Journal</u>	<u>Title of the paper</u>	<u>Title of the proceedings of the conference</u>	<u>Name of the conference</u>	<u>National / International</u>	<u>Year of Publication</u>	<u>ISBN/ ISSN no. of the publication</u>	<u>Affiliating Institute at the time of publication</u>	<u>Name of the publisher</u>	<u>Co-Author (s) (if any)</u>
	Digitalization would be noteworthy for Education, National Seminar on Emerging Trends in ICT January 30-31, 2017, Organized by School of Science, U.P. Rajarshi Tandon Open University Allahabad, 211021								

(d) Seminar/Conference Presentations

A. International

1. 5th International Conference on Soft Materials (ICSM 2022)
Paper Title-MXene-hydrogels and their derivatives applications in an energy storage, sensing, optoelectronics, catalysis and biomedicine, MNIT-Jaipur
Soft Materials Lab, Department of Physics, Malaviya National Institute of Technology Jaipur, India
 11-16 ec. 2022 6 Days
 - 2 “Toward Efficient Drug Delivery and Bio-Imagine through Metal Organic Frameworks”
 International Conference On Biological Research and Bio-Statistics(ICBRBS-2019)
 29-30 April, 2019, organized by *School of Sciences*, U. P. Rajarshi Tandon Open University, Allahabad
 3. “Recent advances of MOFs in drug delivery: A review”
Dinesh Kumar Gupta
 International Conference On *Biological Sciences and Bio-Statistics (ICBS-2017)*, 09-10 March, 2017, organized by *School of Sciences*, U. P. Rajarshi Tandon Open University, Allahabad
 4. “Effect Of Thickness on Optical and Photocatalytic Properties of Nb₂O₅ Thin Films Prepared By Sol- Gel Method”
 Mohd. Danish, Dinesh Kumar Gupta and Ashutosh Pandey
 “International Conference on *Current Scenario and Future Prospects of Nanotechnology and Bio-Statistics (ICNB-2016)*, 25-26 February, 2016, organized by *School of Sciences*, U. P. Rajarshi Tandon Open University, Allahabad
 5. “Green Synthesis of two novel Metal-Organic frameworks”
 Ashutosh Pandey, Sadhana Singh, Dinesh K. Gupta
 2nd International Indo German Symposium, on 29-31 Oct., 2012 organized by Institute of Chemical Technology and Leibniz Institute for Catalysis, at Mumbai, India
 6. “An Efficient Synthesis of Biologically Active Pyrazine Linked Novel Hydrazones”
 G.Thiyagarajan, P.Nithya, Dinesh Kumar Gupta, Ashutosh Pandey
 “International Conference on Biologically Active Molecules (ICBAM-2012)”
 Gandhigarm Rural Institute- Dindigul, Tamil Nadu on 9-11 March, 2012
 ISBN- 93-82062-03-3
 7. “Synthesis and Characterization of Metal Oxides Nanomaterials using modified alkoxides as Precursor”
Dinesh Kumar Gupta and Ashutosh Pandey
 International Conference on Advanced Nanomaterials and Nanotechnology (ICANN-2009)”
 during 09-11 Dec. 2009, Organized by Center for Nanotechnology, IIT-Guwahati, India.
- B. National**
1. Dr. Dinesh K Gupta
National Seminar on “Nanomaterials and Green Energy”
Department of Chemistry, C.M.P. Degree College, University of Allahabad, Prayagraj.
20th April, 2023
1 day
 2. “Advances of software in exploring Chemistry”
Dinesh Kumar Gupta
 National Seminar On Applications of Scientific and Statistical Software in Research, 30-31 March, 2017, organized by School of Sciences, U.P. Rajarshi Tandon Open University, Allahabad, U.P., India
 3. “Digitalization would be noteworthy for Education”
Dinesh Kumar Gupta
 National Seminar on Emerging Trends in ICT
 January 30-31, 2017, Organized by School of Science, U.P. Rajarshi Tandon Open University Allahabad, 211021
 4. “Green ICT with Green Social Work”
Dinesh Kumar Gupta

National Seminar on Green Social Work on 11th March 2015 by School of Social Sciences, U. P. Rajarshi Tandon Open University, Allahabad

5. **“Role of Science and Technology with management in the Development of Smart City”**
Dinesh Kumar Gupta and Nishtha Gupta
“National Seminar on Technology & Management of Smart Cities: Issues and Opportunities” organized by School of Management Studies, U P Rajarshi Tandon Open University, Allahabad on 3rd May 2015.
6. **One day Workshop on “Distance Education Community College” on 24 November, 2014,**
by U P Rajarshi Tandon Open University, Allahabad
7. **Two day workshop on “Development of Self Learning Material” Organized by**
U P Rajarshi Tandon Open University, Allahabad on 12-13 February, 2014.
8. **“Green Chemistry is an approach for Sustainable Development”**
“National Seminar on Sustainable production of Nutritive Food” Organized by
U P Rajarshi Tandon Open University, Allahabad on 11-12 December, 2013.
9. **“Short term course on Soft Strategies for the Scalable Synthesis of Nanomaterials” during**
27-31 May, 2013. Organized By Department of Chemistry, Motilal Nehru National Institute of
Technology, Allahabad, Allahabad, India
10. **“National Seminar on Science and Technology- Inclusive Innovation” during 5-6 Nov.2011,**
The Allahabad Chapter of Indian Science Congress Association (ISCA)
Organized by Department of Chemistry, University of Allahabad, Allahabad.
11. **“National Symposium-cum-Workshop on X-ray Crystallography” during 08-09 March**
2010, at Department of Chemistry, Banaras Hindu University, Varanasi, India,
Dinesh K Gupta, MNNIT, Allahabad.
12. **“Recycling Helps to Maintain Soil Fertility”,**
Dinesh Kumar Gupta,
7th Indian Agricultural Scientists & Farmer’s Congress, 19-20 Feb., 2005
at Sardar Ballabh Bhai Patel University of Agriculture & Technology, Meerut, India
Organized by Bioved Research & Communication Center, Allahabad, India.

(e) Resource Lectures Delivered

Programme name	Programme Code	Name of electronic media and other digital components incorporated	Year	Link to the relevant portion of documents where electronic media and other digital components have incorporated
B.Sc.	UGCHE-101, Inorganic Chemistry I (Basic Inorganic Chemistry) Unit-1, Atomic Structure	Video Lectures	uploaded on 15.11.2019,	https://youtu.be/aHtF7_3YmuQ

B.Sc.	UGCHE-101, Inorganic Chemistry I (Basic Inorganic Chemistry) Unit-2, Module-1, Periodic Properties of Elements	Video Lectures	uploaded on 06.10.2020,	https://youtu.be/sGfiD9
B.Sc.	UGCHE-101, Inorganic Chemistry I (Basic Inorganic Chemistry) Unit-2, Module-2, Periodic Properties of Elements	Video Lectures	uploaded on 13.08.2021,	https://youtu.be/ KZyeo6jNog
B.Sc.	UGCHE-101, Inorganic Chemistry I (Basic Inorganic Chemistry) Unit-2, Module-3	Video Lectures	uploaded on Mar 7, 2022,	https://www.youtube.com/watch?v=1BY6a1It9G0

(f) Seminars/Conferences/Workshops Organized

- 1. International Conference On Biological Research and Bio-Statistics (ICBRBS-2019)
29-30 April, 2019, organized by School of Sciences, U. P. Rajarshi Tandon Open University, Prayagraj.
U. P. India**
- 2. International Conference on Biological Sciences and Bio-Statistics (ICBS-2017),
Capacity -Organizing Secretary
09-10 March, 2017, organized by School of Sciences, U. P. Rajarshi Tandon Open University, Allahabad**
- 3. “International Conference on Current Scenario and Future Prospects of Nanotechnology and Bio-Statistics (ICNB-2016),
Capacity- Organizing Secretary
25-26 February, 2016, organized by School of Sciences, U. P. Rajarshi Tandon Open University, Allahabad**

(g) Public Service / University Service / Consulting Activity

N/A

(h) Memberships of Academic/Professional Bodies

- 1. Member/Secretary of Board of Studies, Subject-Chemistry, U.P. Rajarshi Tandon Open University, Allahabad, Since 2013 to Till date**
- 2. Reviewer: Applied Biochemistry and Biotechnology (An International Peer Reviewed Research Journal), Springer International Publishing, Since 2015 to Till date**
- 3. Editorial Board Member: Advanced Scientific Research, Allahabad (An International Peer Reviewed Research Journal Of Science And Technology) (ISSN: 2456-7744) Since 2015 to Till date**

(i) <u>Articles</u> N/A							
(j) <u>Professional Development Programme viz Orientation, Refresher, FDP attended (Title, duration, University/Institute where attended)</u>							
	Name of the faculty	Year	FDP/ STTP/ Workshop Title	Organized by	Sponsored/ Approved/ Financed by	Dates	Duration in days
	Dr. Dinesh K Gupta	2023	International Training-cum-Workshop on "Integration of GIS and Disaster Risk Reduction for a Resilient and Sustainable Community	Department of Environmental Science, Integral University, Lucknow	State Disaster Response Force (SDRF) Uttar Pradesh.	29 Nov-5 Dec, 2023	7 days
	Dr. Dinesh K Gupta	2023	Faculty Development Program on " Recent Advances in Chemical and Molecular Science "	Amity Institute of Click Chemistry Research and Studies, Amity University Noida	Amity Institute of Click Chemistry Research & Studies	31 July – 4 Aug, 2023	5 days
	Dr. Dinesh Kumar Gupta	2022	Workshop on Capacity Building Programme on Online Content Development Using OER	Center for Internal Quality Assuranc, U.P. RAJARSHI TANDON OPEN UNIVERSITY, PRAYAGRAJ-U.P.	COMMONWEALTH EDUCATIONAL MEDIA CENTRE FOR ASIA, NEW DELHI In collaboration with U.P. RAJARSHI TANDON OPEN UNIVERSITY, PRAYAGRAJ-U.P.	19-21 July 2022	3 days
	Dr. Dinesh Kumar Gupta	2022	IPR- Policy and Procedure	Research and Development Cell, RAJARSHI TANDON OPEN UNIVERSITY, PRAYAGRAJ-U.P.	U.P. RAJARSHI TANDON OPEN UNIVERSITY, PRAYAGRAJ-U.P.	2 July 2022	1 day
	Dr. Dinesh Kumar Gupta	2022	Teachers Orientatation Programme for Capacity Building on Satbility of Mind and Emotional Wellness	School of Education, UP RAJARSHI TANDON OPEN UNIVERSITY, PRAYAGRAJ	UP RAJARSHI TANDON OPEN UNIVERSITY, PRAYAGRAJ-U.P.	9 may 2022	1 day
	Dr. Dinesh Kumar Gupta	2022	Symposium on Data for Sustainable Development	School of Science, UP RAJARSHI TANDON OPEN UNIVERSITY, PRAYAGRAJ	NSSO, Govt of India and UP RAJARSHI TANDON OPEN UNIVERSITY, PRAYAGRAJ	29 June 2022	1 day
	Dr. Dinesh Kumar Gupta	2022	Online Workshop on Adoption of Open Educational	Center for Internal Quality Assuranc, U.P. RAJARSHI	COMMONWEALTH EDUCATIONAL MEDIA CENTRE FOR ASIA, NEW	17-19 Jan, 2022.	3 days

			Resources in UPRTOU: <i>A Capacity Building Programme</i>	TANDON OPEN UNIVERSITY, PRAYAGRAJ-U.P.	DELHI In collaboration with U.P. RAJARSHI TANDON OPEN UNIVERSITY, PRAYAGRAJ-U.P.			
	Dr. Dinesh Kumar Gupta	2022	<i>Online Course on Understanding Open Educational Resources (OER)</i>	COMMONWEALTH EDUCATIONAL MEDIA CENTRE FOR ASIA, NEW DELHI	NA	19 Jan 2022		

Projects (With Title, Year, Grants, Funding Agency and Collaborations)

Sr. No.	Name of the Project/ Chairs	Name of the Principal Investigator / Co-investigator	Name of the funding agency	Type (Government /Non-Government)	Department /School of Principal Investigator /Co-investigator	Year of award	Funds provided (INR in rupees)	Duration of the project
1	Seasonal Changes in Holi River Ganga: Water Quality Analysis and Environmental Impact	Dr. Sanjay K Singh(PI) And Dr. Dinesh K Gupta (Co-PI)	U.P. State Govt	U.P. State Govt. 10 Nov. 2022	School of Social Science	2022	2,17,800.00	On-going

Administrative Positions/Assignments Held

- 1. Member/Secretary of Board of Studies, Subject-Chemistry, U.P. Rajarshi Tandon Open University, Allahabad, Since 2013 to Till date**

Academic Foreign Visits

N/A

Any Other Details

N/A

Signature of Faculty Member
(in Hard Copy Only)

Note: Please submit this file to E-mail ID: iqacuprtou@gmail.com through your own E-mail ID. Also submit the hard copy of the profile to the Director IQAC

UNIVERSITY OF ALLAHABAD



Master of Science

Enrolment No. 00010263

Roll No. 13088

This is to certify that Dinesh Kumar Gupta
Son/Daughter of Sri Ram Anwar Gupta Mahuley
 obtained the degree of Master of Science in this University
 in the Examination of 2005; in Chemistry; and
 that he/she was placed in the First class.

University of Allahabad:

The 18th Oct, 2006
 The 14th Dec, 2007.

Rajesh Kumar
 Vice-Chancellor

मोतीलाल नेहरू राष्ट्रीय प्रौद्योगिकी संस्थान इलाहाबाद

विद्या परिषद् की अनुशंसा पर

दिनेश कुमार गुप्ता

को

अगस्त २०१२ में शोध-प्रबंध के सफल पक्षपोषण के उपरान्त

डॉक्टर ऑफ़ फिलॉसफ़ी

की संस्थान मुद्रा अंकित उपाधि समस्त सम्मानों, विशेषाधिकारों व दायित्वों के साथ आज २५ दिसम्बर २०१२ को इलाहाबाद (भारत) में प्रदान करता है

Motilal Nehru National Institute of Technology Allahabad

*on the recommendation of the Senate
hereby confers on*

Dinesh Kumar Gupta

*on successful defence of the thesis in August 2012
the Degree of*

Doctor of Philosophy

*with all Honours, Privileges and Obligations thereunto pertaining on this
day the 25th December 2012, under Seal of the Institute at Allahabad (INDIA).*

**Thesis Title: Synthesis and Characterization of Some Modified Metal Alkoxides As Precursors for
Metal Oxide Nanomaterials and Some Metal Organic Framework Compounds.**



Chairperson, Board of Governors

अध्यक्ष, प्रशासकीय परिषद्

Chairperson, Board of Governors

Chairman, Senate

अध्यक्ष, विद्या परिषद्

Chairman, Senate

ई-डिस्ट्रिक्ट के जनसर्जन जारी.



उत्तर प्रदेश शासन

FORM OF CERTIFICATE TO BE PRODUCED BY OTHER BACKWARD CLASSES
APPLYING FOR APPOINTMENT TO POSTS UNDER THE GOVERNMENT OF INDIA

जिला लखीमन
सदर 231750030190916
आवेदन क्र० 453233020499
प्रमाणपत्र क्र०

जारी दिनांक: 05/08/2023



This is to certify that DINESH KUMAR GUPTA son/daughter of LATE RAM AVTAR GUPTA MAHULEY mother's name TULSA DEVI R/o B-42, GOVINDPUR COLONY Tehsil सरत District प्रयागराज in the Uttar Pradesh state belongs to the Habasi Community which is recognized as a backward class under the Government Of India, Ministry of Welfare Resolution No. 12011/68/93-BOC(C) dated 10th Sept. 1993, published in the Gazette of India Extra Ordinary Part-I Section-I Dated 13th Sept. 1993 and onwards till date.

DINESH KUMAR GUPTA and/or his family ordinarily reside (s) in the B-42, GOVINDPUR COLONY of the Tehsil सरत District प्रयागराज of the Uttar Pradesh state.

This is also to certify that he/she does not belongs to the persons/sections (Creamy Layer) mentioned in column 3 of the schedule to the Government Of India, Department of Personnel & Training O.M.No. 36012/22/93 Estt(SCT) dated 08-09-93 or the latest notification of the Government of India, which is modified vide OM No. 36033/3/2004 Estt.(Res.) dated 09/03/2004 and further modified vide OM No. 36033/3/2004-Estt. (Res.) dated 14/10/2008 or the latest notification of the Government of India.



जारी कर्ता केन्द्र: गंजीत नुगार प्रयागराज, जाली कम्युनिभेचनRAVINDRA
पद: गंजीत कुमार आपसवाल, केन्द्र प्रजारी
स्थान: मास्टर हसन रोड निगर बैंक ऑफ़ थनोदा सटरा
जांच, सदर Allahabad

दिनांक: 19/08/2023

हस्ताक्षर निच सदर

Duly
Signed by
RAVINDRA
PRITAP
ON-PERSONAL
SINGH
Pradesh

सहायक अधिकारी/कहासीलदार
डिस्ट्रिक्ट हस्ताक्षरित
सदर, प्रयागराज

दिनांक: 05/08/2023

यह प्रमाण पत्र को प्रमाणित करने के लिए डिस्ट्रिक्ट सिविल सेशन कोर्ट में जमा किया गया है तथा डिस्ट्रिक्ट सिविल सेशन कोर्ट के जज द्वारा जारी किया गया है।
यह प्रमाण पत्र को प्रमाणित करने के लिए डिस्ट्रिक्ट सिविल सेशन कोर्ट में जमा किया गया है तथा डिस्ट्रिक्ट सिविल सेशन कोर्ट के जज द्वारा जारी किया गया है।

Signature



कुलसचिव

उ.प्र. राजर्षि टण्डन मुक्त विश्वविद्यालय
विश्वविद्यालय परिसर, शान्तिपुरम (सेक्टर -एफ). फाफामऊ
प्रयागराज - 211013

मो. नं. : 7525048031

ई-मेल registrar.uprtou@gmail.com

दिनांक:- 23-03-2022

जिस किसी से सम्बन्धित हो

प्रमाणित किया जाता है कि डॉ. दिनेश कुमार गुप्ता, उत्तर प्रदेश राजर्षि टण्डन मुक्त विश्वविद्यालय, प्रयागराज में सहायक आचार्य (संविदा) (रसायन विज्ञान) के रूप में नियत मानदेय पर संविदा के आधार पर दिनांक-20/02/2013 से कार्यरत है। उक्त पद पर डॉ. गुप्ता की नियुक्ति विश्वविद्यालय के परिनियम बिन्दु संख्या 3.19 (ii) में दी गयी व्यवस्थाओं एवं अनुबन्ध की शर्त के बिन्दु-08 के क्रम में की गयी है। डॉ. गुप्ता को माह फरवरी-2022 का इस विश्वविद्यालय द्वारा रु. 33,000/- प्रतिमाह नियत मानदेय की दर से भुगतान किया गया है।

मैं इनके उज्ज्वल भविष्य की कामना करता हूँ।


प्रो. प्रेम प्रकाश दुबे
कुलसचिव

049 / 3912201

हाई स्कूल परीक्षा-अंक पत्र

1283917

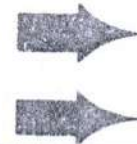
माध्यमिक शिक्षा परिषद उत्तर प्रदेश, द्वारा संचालित वर्ष १९९८ की हाई स्कूल परीक्षा में निम्नलिखित परीक्षार्थी द्वारा प्राप्त अंकों का विवरण :

अनुक्रमांक परीक्षार्थी का नाम जन्मतिथि क्रमांक
 0894367 DINESH KUMAR GUPTA 25/09/82 3901797

विद्यालय/केन्द्र का नाम परीक्षा प्रवर्ग, संस्था/व्यक्ति
 BHARAT SCOUTS & GUIDE HS SCH ALLAHABAD FULL EXAM REGULAR

विषय	अधिकतम अंक	प्रश्न-पत्रवार प्राप्तांक	योग	सम्पूर्ण योग एवं परीक्षा फल
HINDI	100	1/23 2/16 3/22	061	353
ENGLISH	100	1/38 2/26	064	PASSED
MATHEMATICS TWO	100	1/35 2/18	053	SECOND DIV
SCIENCE TWO	100	1/19 2/17 36 P/18	054	
SOCIAL SCIENCE	100	1/23 2/27	050	CAT ME-B
BIOLOGY	100	1/28 2/26 54 P/17	071	

जांचकर्ता के हस्ताक्षर  प्रधानाचार्य
 दिनांक 16/7/98 के हस्ताक्षर
 नोट-आवश्यक सूचना पीछे मुद्रित है। मुहर




 Principal
 Bharat Scouts & Guides
 H.S. School, Allahabad

418830

39/049/3906560

अनुक्रमांक

0894367

मानविक शिक्षा परिषद्, उत्तर प्रदेश
 मानविक शिक्षा परिषद्, उत्तर प्रदेश
 मानविक शिक्षा परिषद्, उत्तर प्रदेश



हाई स्कूल परीक्षा, १९९८

HSIBUP - 1998

प्रमाणित किया जाता है कि परिषद् के अभिलेखानुसार

DINESH KUMAR GUPTA

आत्मज श्री

RAM AUTAR GUPTA MAHULEY

ने जिनकी जन्मतिथि

25TH SEPTEMBER NINETEEN HUNDRED EIGHTY TWO (25-09-82)

है, मार्च/अप्रैल १९९८ की हाईस्कूल परीक्षा निम्नलिखित विवरण के अनुसार उत्तीर्ण की है -

1-HINDI	2-ENGLISH	3-MATHEMATICS TWO
4-SCIENCE TWO	5-SOCIAL SCIENCE	6-BIOLOGY
CATEGORY OF MORAL EDUCATION-B		

SECOND

उत्तीर्ण श्रेणी

विद्यालय/केन्द्र

BHARAT SCOUTS & GUIDE HS SCH ALLAHABAD

प्रमाण पत्र क्रमांक

इलाहाबाद

दिनांक

3900704

REGULAR

23rd JUNE, 1998.

(अचला खन्ना)

(अचला खन्ना)
 सचिव

'D' का अर्थ सम्बन्धित विषय में 'विशेष योग्यता' है।
 'HONOURS' का अर्थ 'सम्मान सहित उत्तीर्ण' है।
 दीप-विषय का पूर्ण विवरण पृष्ठ भाग पर अंकित है।

इन्टरमीडिएट परीक्षा-अंक पत्र

0857053

माध्यमिक शिक्षा परिषद् उत्तर प्रदेश, द्वारा संचालित वर्ष २००० की इन्टरमीडिएट परीक्षा में निम्नलिखित परीक्षार्थी द्वारा प्राप्त अंकों का विवरण :

अनुक्रमांक

परीक्षार्थी का नाम

वर्ग

क्रमांक

432056

DINESH KUMAR GUPTA

B1

I6300975

विद्यालय/केन्द्र का नाम

परीक्षा प्रवर्ग,

63/001/0975
संस्था/व्यक्ति

GOVT INT COLLEGE ALLAHABAD

FULL EXAM

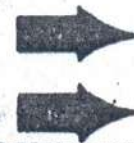
REGULAR

विषय	अधिकतम अंक	प्रश्न-पत्रवार प्राप्तांक	योग	सम्पूर्ण योग एवं परीक्षाफल
GENERAL HINDI	100	1/ 17 2/ 13 3/ 18	048	243/ 500
ENGLISH	100	1/ 25 2/ 17	042	PASSED
PHYSICS	100	1/ 12 2/ 15 027	P/ 16 043	SECOND DIV
CHEMISTRY	100	1/ 20 2/ 21 041	P/ 13 054	
BIOLOGY	100	1/ 15 2/ 19 034	P/ 22 056	

जांचकर्ता के हस्ताक्षर
दिनांक

[Signature]
1.7.2000

प्रधानाचार्य
के हस्ताक्षर



नोट - आवश्यक सूचना पीछे मुद्रित है।

मुहर

प्रधानाचार्य

राजकीय इंटर कॉलेज

इलाहाबाद

579160

अनुक्रमांक
0432056

मानविकी का आर्य समाज विश्व शिक्षा परिषद्, उत्तर प्रदेश इण्टरमीडिएट एजुकेशन यू.पी.



HSIBUP-2000

इण्टरमीडिएट परीक्षा, २०००

प्रमाणित किया जाता है कि परिषद् के अभिलेखानुसार

DINESH KUMAR GUPTA

आत्मज/आत्मजा श्रीमती TULSA GUPTA

एवं श्री

SRI RAM AVTAR GUPTA MAHULAY

ने वर्ष २००० की

इण्टरमीडिएट परीक्षा निम्नलिखित विवरण के अनुसार उत्तीर्ण की है -

1. GENERAL HINDI

2. ENGLISH

3. PHYSICS

4. CHEMISTRY

5. BIOLOGY

उत्तीर्ण श्रेणी

SECOND

विद्यालय/केन्द्र

GOVT INT COLLEGE ALLAHABAD

प्रमाण पत्र क्रमांक

इलाहाबाद

दिनांक

I6301991

REGULAR

24 JUNE, 2000

63/001/0802

'D' का अर्थ सम्बन्धित विषय में 'विशेष योग्यता' है।
'HONOURS' का अर्थ 'सम्मान सहित उत्तीर्ण' है।
टीप-विषय का पूर्ण विवरण पृष्ठ भाग पर अंकित है।

(अचला खन्ना)
सचिव

UNIVERSITY OF ALLAHABAD

Statement of Marks of B.Sc. Part III Examination, 2003
(Regular Candidate of Three Year Degree Course)

Name : DINESH KUMAR GUPTA R. No: 525056
Father's Name: RAM AVTAR GUPTA MAHULEY E. No: U0010263

Subject	Theory				Total	Pract.	Subject	
	1st	2nd	3rd	4th			Total	GM
Botany	24	36	33		93/150	60/75	153/225	
Chemistry	25	21	30		76/150	64/75	140/225	

Pt. I: 254 (2001) Pt. II: 277 (2002) Part III : 293/450

Grand Total : 824/1350

Result First Division

To pass, a candidate must obtain a minimum
of 33% marks in theory as well as in practical
of each subject separately.

Division : First 810, Second 648, Third 450.

MFEAS8961

A.K. Srivastava

June 18, 2003

Controller of Examinations

UNIVERSITY OF ALLAHABAD



Bachelor of Science

Enrolment No. 2100102263

Roll No. 525056

This is to certify that Dinesh Kumar Gupta
Son/Daughter of Sri/Smt. Ram Arjun Gupta Mahuley
of University of Allahabad
obtained the degree of Bachelor of Science in this University
in the Examination of 2003; and that he/she was placed
in the First class.

The subjects in which he/she was examined were:

B. Sc. Part I and B. Sc. Part II

1. Botany
2. Chemistry
3. Zoology

B. Sc. Part III

1. Botany
2. Chemistry

University of Allahabad:

The 7th, July, 192006.

Rajendra Kumar
Vice-Chancellor.

UNIVERSITY OF ALLAHABAD

M.Sc. (Chemistry) Final Examination, 2005

Name : DINESH KUMAR GUPTA
Father's Name : RAM AUTAR GUPTA MAHULEY

Roll No.: 13088

Paper:	Subject - Inorganic	MARKS
I-Theoretical Inorganic Chemistry		47/75
II-Inorganic Chemistry		48/75
III-Coordination Chemistry and Bio-Inorganic Chemistry		50/75
IV-Elective Paper E		44/75
	Theory Total	189/300
	Practical	129/150
Previous : 306/450	Final : 318/450	

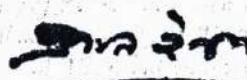
Result : I Division

Grand Total : 624/900

To Pass a candidate must obtain at least 36% marks in Theory as well as in Practical separately.

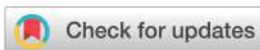
DIVISION: I(60%); II(48%); III(36%)

September 02, 2005


Controller of Examinations

582376 005850620

REVIEW



Cite this: *J. Mater. Chem. B*,
2024, 12, 2691

Received 29th October 2023,
Accepted 8th February 2024

DOI: 10.1039/d3tb02562d

rsc.li/materials-b

MOF magic: zirconium-based frameworks in
theranostic and bio-imaging applications

Dinesh K. Gupta,^a Santosh Kumar^{*b} and Mohmmad Younus Wani^c

Over the past two decades, metal–organic frameworks (MOFs) have garnered substantial scientific interest across diverse fields, spanning gas storage, catalysis, biotechnology, and more. Zirconium, abundant in nature and biologically relevant, offers an appealing combination of high content and low toxicity. Consequently, Zr-based MOFs have emerged as promising materials with significant potential in biomedical applications. These MOFs serve as effective nanocarriers for controlled drug delivery, particularly for challenging antitumor and retroviral drugs in cancer and AIDS treatment. Additionally, they exhibit prowess in bio-imaging applications. Beyond drug delivery, Zr-MOFs are notable for their mechanical, thermal, and chemical stability, making them increasingly relevant in engineering. The rising demand for stable, non-toxic Zr-MOFs facilitating facile nanoparticle formation, especially in drug delivery and imaging, is noteworthy. This review focuses on biocompatible zirconium-based metal–organic frameworks (Zr-MOFs) for controlled delivery in treating diseases like cancer and AIDS. These MOFs play a key role in theranostic approaches, integrating diagnostics and therapy. Additionally, their utility in bio-imaging underscores their versatility in advancing medical applications.

1. Introduction

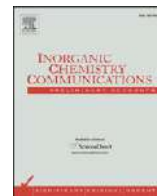
Metal–organic frameworks (MOFs) have become increasingly popular in various applications, ranging from gas storage, purification, separation, catalysis, nonlinear optics, magnetism, to biotechnology.^{1–5} These versatile materials have demonstrated their potential as drug delivery vehicles, joining the ranks of other nanocarriers such as nanoparticles,^{6–8} nanoemulsions,⁹ hydrogels,¹⁰ micelles,^{11,12} and liposomes.^{13,14} The need for drug carriers arises due to limitations associated with the direct delivery of conventional diagnostic and therapeutic drugs. These limitations include poor solubility, rapid clearance, inadequate pharmacokinetics, high dosage requirements, increased side effects, and non-specific targeting. Limited solubility in biological fluids is one of the key challenges encountered with conventional drugs, which hampers their effective delivery. Rapid clearance from the body further diminishes their therapeutic efficacy. Poor pharmacokinetics, which encompasses drug absorption, distribution, metabolism, and excretion, can also impede their effectiveness. Another drawback is the requirement of high dosages of conventional

drugs due to their limited specificity and targeting abilities. Inefficient drug delivery necessitates elevated drug concentrations to achieve the desired therapeutic response, often resulting in heightened side effects and potential harm to healthy tissues. To overcome these limitations, innovative drug delivery systems have been developed, including nanocarriers like Zr-based-MOFs. By encapsulating drug molecules within these carriers, various challenges associated with conventional drugs can be addressed. Nanocarriers offer advantages such as improved solubility, prolonged circulation time, controlled release, targeted delivery, reduced side effects, and enhanced pharmacokinetic profiles. While each type of nanocarrier has its own advantages and disadvantages, the overall objective remains consistent: to provide an optimal platform for drug delivery that surpasses the limitations of conventional drugs. Continued research and innovation in the field of drug delivery aim to revolutionize medical treatments, maximizing therapeutic potential and minimizing the drawbacks associated with traditional drug administration. This comprehensive review article delves into the transformative role of Zr-based MOFs, in addressing the inherent limitations ranging from poor solubility to non-specific targeting of conventional drug delivery methods, and emphasizes the critical role that Zr-based MOFs play in overcoming these hurdles. This review would serve as a valuable resource in advancing our understanding of Zr-based MOFs as an integral component in the ongoing quest for more effective and advanced frameworks for biomedical applications.

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Hydrothermal synthesis, structure and photoluminescent property of a novel sodium-pyrazinyltetrazole framework compound

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Pyrazine tetrazolate (pzta)

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ABSTRACT

A novel coordination polymer of sodium $[\text{Na}_2(\text{Pzta})_2(\text{H}_2\text{O})_2]_n$ (**1**) has been synthesized hydrothermally by reacting pyrazine carbonitrile and sodium azide in the presence of aluminum(III) chloride as Lewis acid catalyst. Complex **1** was characterized by elemental analysis, thermal gravimetric analysis and IR spectroscopy. Single crystal X-ray diffraction revealed that complex **1** has both distorted trigonal bipyramidal and square pyramidal coordination geometries around sodium atoms. Complex **1** represents the first hydrothermally synthesized 1D sodium monotetrazole coordination polymer. Strong hydrogen bonds are formed between water molecules and nitrogen atoms of both pyrazine and tetrazole rings which result in the formation of a 2D metallosupramolecular network. The luminescence properties of **1** were also investigated at room temperature in the solid state.

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Owing to their unique structures and their potential applications in advanced materials [1] coordination compounds with tetrazole-based ligands have been the subject of intense research in recent years. In the earlier literature tetrazoles have reportedly been synthesized by introducing expensive and toxic metals and these methods suffer from problems such as severe reaction conditions and risks of explosions [2]. Champness and Schroder in 1997 and Sharpless and co-workers in 2001 described a safe, convenient, and environment friendly synthetic route to 5-substituted 1H-tetrazoles with water as solvent and a zinc salt as catalyst [3,4]. In contrast to direct tetrazole-synthesis in situ hydrothermal methods showed a [2 + 3] cycloaddition of azides with nitriles along with several advantages like (a) they take place in one step from the reactants (b) no need for separate synthesis of ligands (c) environment friendly and (d) sufficient growth for single crystals [5].

Single crystals of the coordination polymers under hydrothermal conditions have been successfully isolated by a number of methods as the intermediates of these reactions which are helpful in understanding the mechanism of the Demko–Sharpless reaction. Up to now the in situ solvo/hydrothermal approach [6–12] has been extended to metals like Zn(II), Cu(I), Cd(II), Ag(I), Pb(I), Hg(II) etc. with cyanopyridines, 2-amino-5-cyanopyridine, MeCN, cyanobenzene, 4-methylbenzonitrile, malononitrile and dicyanopyridine etc. as ligands. Most of the coordinated metal centers in these compounds are derived from the used Lewis acids during the reactions. In the present synthesis we have used AlCl_3 as a catalyst which is not only a strong Lewis acid but also

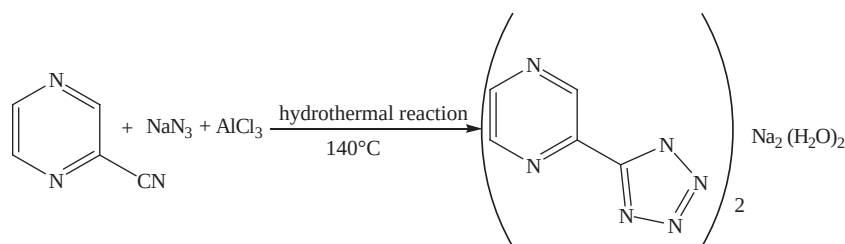
known for lowering the reaction barrier [13]. Surprisingly, the resulting framework structure has been generated around sodium centers instead of aluminum. The first hydrothermally synthesized sodium ditetrazole coordination polymer was reported by Huang et al. [14] by the hydrothermal treatment of ZnCl_2 with NaN_3 and 1,4-dicyano butylenes in the presence of NaOH wherein all the sodium centers have distorted octahedral geometry around them. A mixed metal coordination polymer $[\text{CuNa}_2(\text{tza})_2(\text{H}_2\text{O})_4]_n$ [15], having six coordination around sodium atoms has also been reported.

In this article we report the synthesis, crystal structure, photoluminescence and thermal behavior of a hydrothermally synthesized sodium tetrazole complex, Diaquabis[5-(2-pyrazinyl)tetrazolato] disodium (I) (**1**). Complex **1** constitutes the first example of a sodium tetrazole complex with five coordination around the metal centers in both trigonal bipyramidal and square pyramidal geometries. Compound **1** was synthesized by the reaction of AlCl_3 , 2-Cyanopyrazine, and NaN_3 under hydrothermal condition at 140 °C [16] (Scheme 1). The reaction involved in situ ligand transformation of 2-Cyanopyrazine via cycloaddition reaction with the azide ion to give the 5-(2-pyrazinyltetrazole) ligand. In contrast to the other similar reported reactions, in complex **1**, the metal involved in coordination has not been taken from the used Lewis acid and a novel framework has been framed around sodium atoms which are derived from the used sodium azide.

In the IR spectrum of complex **1** a peak over 3100 cm^{-1} indicated the presence of coordinated water molecules. Furthermore, the absence of absorptions around 2300 cm^{-1} and $\sim 2100\text{ cm}^{-1}$ suggested the absence of cyano and azide groups while the peaks in the range $1557\text{--}1447\text{ cm}^{-1}$ indicated that the [2 + 3] cycloaddition reaction between cyano group and azide anion has taken place.

* Corresponding author. Tel.: +91 5322271276.

E-mail address: apandey70@yahoo.com (A. Pandey).



Scheme 1.

Single crystal X-ray analysis [17] revealed that complex **1** crystallizes in the triclinic space group $P\bar{1}$. The asymmetric unit consists of two sodium atoms: Na1 exhibits distorted square pyramidal coordination geometry made by three N donors from the two pzta ligands and two oxygens from two water (one bridging and one terminal) molecules, Na2 is coordinated by four N atoms from three pzta groups and by one bridging water molecule, displaying distorted trigonal bipyramidal coordination geometry (Fig. 1).

The Na–O distances are in the range of 2.3289(2)–2.4260(2) Å while the Na–N bond lengths are in the range 2.3992(2)–2.5218(2) Å which are in accordance with previous reports [14]. In **1** the bond angle O–Na1–O is 98.14° and angles N–Na1–N are in the range of 69.53°–137.38° while angles O–Na1–N range from 89.55 to 170.29°. The bond angles O2–Na2–N around Na2 are in range of 89.3° to 106.4° while the angles N–Na2–N fall under the limit of 68.14–158.63°. In the sodium-tetrazolate complex reported by Huang et al. each ditetrazolyethylene acts only as a tridentate bridging ligand while in **1**, one of the two tetrazolylpyrazine moieties acts as a tridentate bridging ligand by coordinating with sodium atoms of two neighboring units i.e. Na1 from one unit and both Na1 and Na2 from second unit. The second tetrazolylpyrazine group is bridging bidentately between the two sodium atoms (both Na2 type) taken from two neighboring units. The above mentioned two types of the bridging coordination of the tetrazolylpyrazine ligand results in the formation of 1D chain. It is noteworthy that the six sodium atoms, two oxygen atoms and eight nitrogen atoms (from four different tetrazolylpyrazine ligands) fabricate a sixteen membered sigmoid ring in 1D chain (Fig. 2).

The most remarkable feature of **1** is the molecular self-organization of the neutral $[\text{Na}_2(\text{Pzta})_2(\text{H}_2\text{O})_2]_n$ building blocks into a 2D open framework (Fig. 2) via intermolecular hydrogen bonding interactions between the non-coordinated nitrogen atoms of the

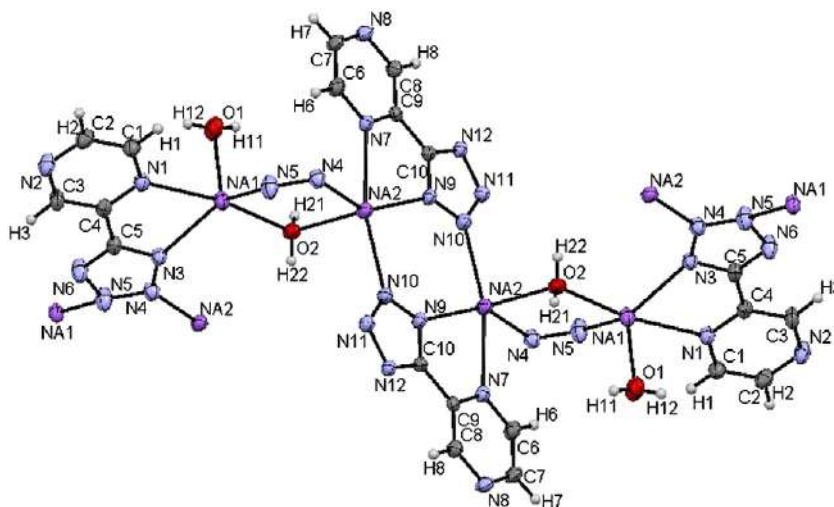
pyrazine and tetrazole ring and coordinated water molecules. The coordinated water molecules act as H-bond donor while pyrazinyl tetrazolate nitrogen atoms behave as acceptors.

Of the three hydrogen bonding sites in the metal-complex building block, the one oriented along the *c*-axis is linked by $\text{wO} \cdots \text{H} \cdots \text{N}$ bonds (between the bridging water molecule and pyrazine ring) via $R^2_2(20)$ molecular units while the another two are formed by the bridging and terminal water molecules which are linked to two different nitrogen atoms of the same tetrazole group via $R^2_2(7)$ molecular units. Formation of both type of H-bonded molecular units results in the establishment of a 2D supramolecular network along the *c*-axis (Fig. 3).

As depicted in Fig. 2, there is a close layer packing of 1D layered loop chain formed by neutral $[\text{Na}_2(\text{Pzta})_2(\text{H}_2\text{O})_2]_n$ repeating units utilizing pyrazinetetrazolyl metal coordination mode accompanied by the bridging ligands. Thus the metallosupramolecular motif in **1** is neutral and due to absence of charged groups the structure is an outcome of neutral–neutral type of H-bonding interaction.

Thus, the molecular units $R^2_2(20)$ and $R^2_2(7)$ appear to have reliable utility for constructing a 2D supramolecular structure from 1D coordination chain.

Compound **1** was analyzed by thermogravimetric analysis (TGA) under inert atmosphere, while ramping the temperature at a rate of $10^\circ\text{C min}^{-1}$ from 0 to 1200°C (Fig. S2). The thermal decomposition profile of **1** indicated that it was stable up to 225°C followed by a rapid weight loss in the temperature range $235\text{--}315^\circ\text{C}$ corresponding to one molecule of water (calc. 4.22%, obs. 4.78%). The weight loss between 315 to 357°C has been attributed to the decomposition of one tetrazolato group (calc. 18.99%, obs. 20.37%). In the temperature range between 357 to 397°C , the loss of second water molecule has taken place (calc. 6.20%, obs. 5.84%). In the temperature range $397\text{--}460^\circ\text{C}$, 1.5 molecules of N_2 was liberated (calc. 7.72%, obs. 9.72%). Finally two pyrazine groups were

Fig. 1. ORTEP diagram (30% probability ellipsoids) showing the coordination atmospheres of Na atoms in **1**.

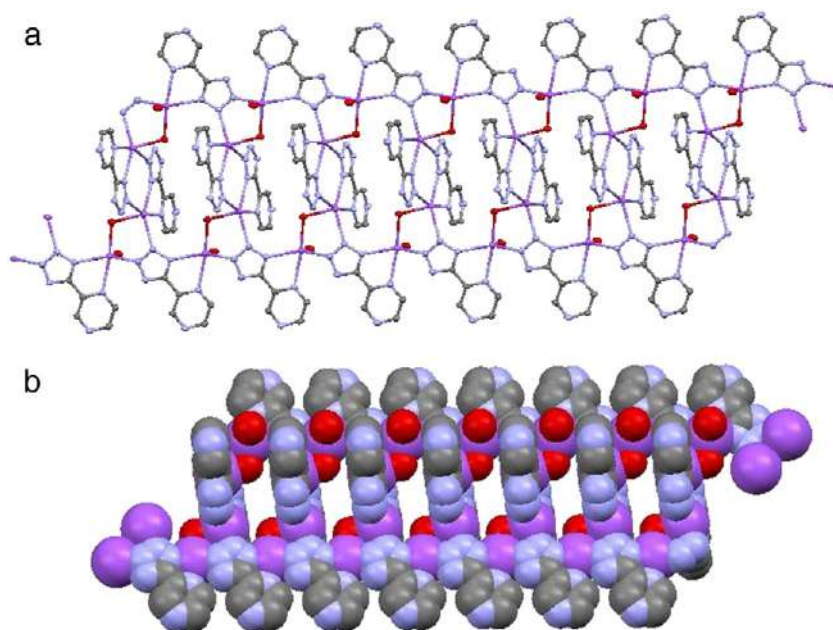


Fig. 2. View of the crystal lattice along the *b*-axis showing: (a) 1D double chain coordination polymer. (b) The space filled diagram showing the sigmoid cavity.

released between 847 to 1088 °C (calc. 82.86%, obs. 80.56%) from the metal complex and the cyano group left on the metal atom was further oxidized to give the sodium oxide residue [12].

Photofluorescent measurements of the complex **1** was carried out in the solid state at room temperature. The free pzta ligand exhibits blue fluorescent emission band at 452 nm. As illustrated in Fig. 4, complex **1** upon excitation at 385 nm shows fluorescent emission band with maximum intensity at 436 nm. Compared with free pzta (452 nm) [18], the emission peak is blue shifted in **1**, which can tentatively be assigned

to the intra-ligand transition of pzta modified by metal coordination. Since compound **1** has high thermal stability and is virtually insoluble in most common solvents such as acetone, methanol, chloroform, benzene, water etc. the emission property makes it a potential photoactive material.

In the present study we have synthesized a novel sodium (I) coordination complex by using AlCl_3 as Lewis acid catalyst. The structure of the complex contains both covalent and non-covalent interactions. The propagation of the low dimensional structure of the sodium

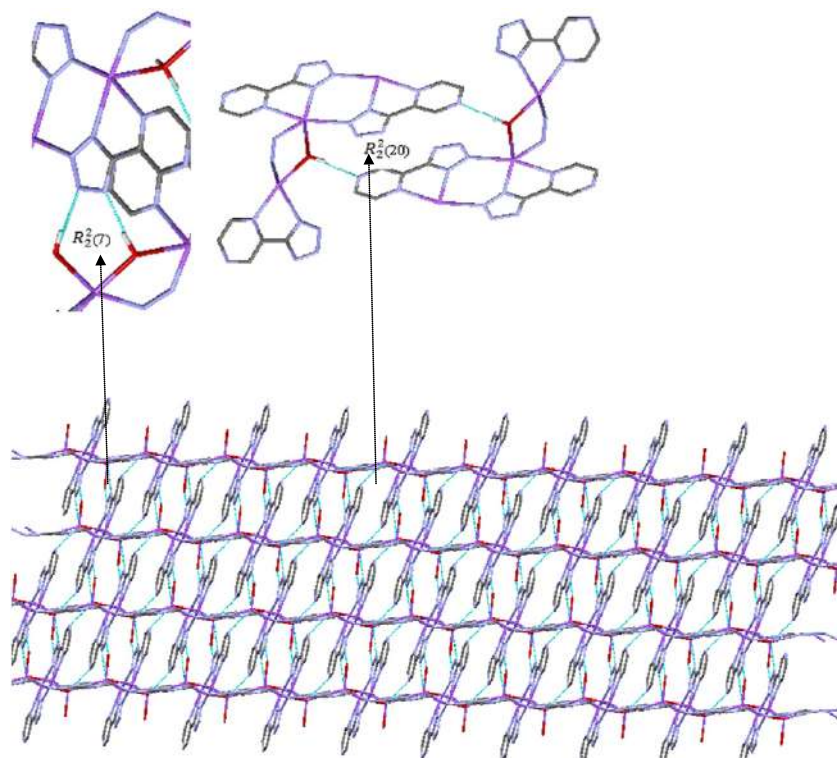


Fig. 3. View of the crystal lattice showing the 2D H-bonded metallo-supramolecular network.

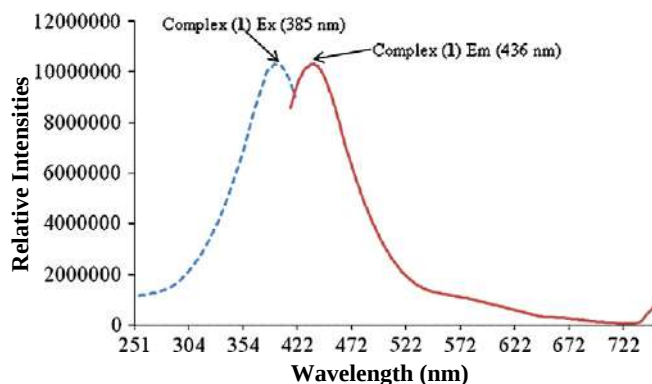


Fig. 4. The solid-state fluorescent spectrum of **1** at room temperature.

complex to 2D metal-organic framework structures results, mainly due to the robust hydrogen bonding interactions involving a variety of molecular units. This work supports Sharpless' cycloaddition reaction of nitriles and azides by using Al(III) in place of Ag(I)/Cu(II/I)/Zn(II)/Cd(II) as catalyst to generate interesting metal coordination polymers. This may also be a new approach for the construction of ligand based coordination polymers containing other main group metals with interesting structural and functional properties by involving the [2 + 3] cycloaddition reaction of an azide with a nitrile.

Acknowledgment

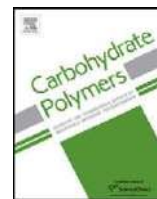
This work was supported by the DST project grant SR/ME/S-3/0016/2008.

Appendix A. Supplementary material

CCDC 809007 contain the supplementary crystallographic data for **1**. This data can be obtained free of charge via <http://www.ccdc.cam.ac.uk/conts/retrieving.html>, or from the Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: (+ 44) 1223-336-033; or e-mail: deposit@ccdc.cam.ac.uk. Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.inoche.2011.05.052.

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- [16] Synthesis of **1** A mixture of AlCl₃ (0.410 g; 3.0 mmol), pyrazinecarbonitrile (0.312 g; 3.0 mmol), NaN₃ (0.195 g; 3.0 mmol), C₂H₅OH (5 ml) and H₂O (10 ml), sealed in a 20 ml Teflon-lined reactor was heated in an oven at 140 °C for 50 h followed by cooling to room temperature at a rate of 5 °C h⁻¹. The resultant solution was allowed to evaporate slowly at room temperature for several weeks to give colourless crystals of **1** (yield: 61%). Elemental analysis (%) calc. for Na₂C₁₀H₁₀O₂: C, 31.78; H, 2.62; N, 44.61. Found: C, 31.89; H, 2.65; N, 44.65. IR (KBr, cm⁻¹): 3441(m), 3048(w), 1623(w), 1540(s), 1427(m), 1340(s), 1280(s), 1180(m), 1155(s), 1070(m), 1018(s), 952(w), 891(m), 771(m).
- [17] Crystal data for complex **1**: triclinic, space group *P*1 with *a* = 6.6586(5) Å, *b* = 7.2459(6) Å, *c* = 16.7473(10) Å, α = 6.6586(5)°, β = 88.072(5)°, γ = 76.940(6)°, *V* = 786.515 Å³, *Z* = 2, *D_c* = 1.5887 g/cm³, *R*(*F*_{obs}) = 0.0343, MoK α radiation, λ = 0.71073, μ = 0.166 mm⁻¹, *R_w*(*F*²) = 0.0764, *R_i* = 0.0269, *wR*₂ = 0.0575 [*I* > 2 σ (*I*)]. X-ray data collection was carried out using a 'Oxford XCalibur' diffractometer with a CCD area detector. The structure was solved by direct methods and refined through full-matrix least-squares techniques using the WINGX, SHELX-97 programs.
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Synthesis of superparamagnetic bare Fe₃O₄ nanostructures and core/shell (Fe₃O₄/alginate) nanocomposites

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ABSTRACT

In this article we report about the synthesis of superparamagnetic bare Fe₃O₄ nanostructures and core/shell (Fe₃O₄/alginate) nanocomposites by simple low-temperature based method at pH values 5, 9, and 14. The structural morphology and magnetic behavior of Fe₃O₄ nanostructures and core/shell (Fe₃O₄/alginate) nanocomposites (Fe₃O₄/alg NCs) have been investigated by X-ray diffractometer (XRD), Fourier transform infrared spectroscopy (FT-IR), Raman spectroscopy (RS), ultraviolet–visible (UV–vis) spectroscopy, transmission electron microscopy (TEM), energy dispersive X-ray spectroscopy (EDX) and vibrating sample magnetometer (VSM). The particle size was calculated by TEM measurements and it turns out to be ~10 nm and ~14 nm for bare Fe₃O₄ nanoparticle and Fe₃O₄/alg NCs with core/shell structure, respectively. The magnetic properties of the synthesized products were found to be function of pH at which the synthesis has been done. The synthesized Fe₃O₄ nanoparticle and Fe₃O₄/alg NCs were found to be superparamagnetic in nature at room temperature. We observed that the value of saturation magnetization in case of Fe₃O₄/alg NCs decreases by increasing the pH value.

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1. Introduction

Currently, the investigations on magnetic nanoparticles (MNPs) have attracted considerable attention to the scientist due to its application in various fields such as physics, medicine, biology and materials science. Particularly, superparamagnetic materials play an important role in biomedical applications including magnetic resonance imaging for clinical diagnosis, magnetic drug targeting, hyperthermia anti-cancer strategy (Gupta & Gupta, 2005; Kim et al., 2008; Qu, Liu, Wang, & Hong, 2010) and enzyme immobilization (Singh, Kalita, Singh, & Malhotra, 2011; Singh, Srivastava, Dutta, & Dutta, 2011). Magnetite (Fe₃O₄) and maghemite (γ-Fe₂O₃) are the most commonly used magnetic carriers for a variety of biomedical applications (Liu et al., 2007). Among the various types of magnetic materials, the magnetic–polymer composites represent a class of functional materials where magnetic NPs are embedded in polymer matrixes. Recently, a considerable number of studies

have been done on magnetic–polymer nanocomposites such as; PVP-coated Fe₃O₄ (Liu et al., 2007), Fe₃O₄–chitosan and α-Fe₂O₃ NPs (Li, Jiang, Huang, Ding, & Chen, 2008; Singh, Kalita, et al., 2011; Singh, Srivastava, et al., 2011), tyrosinase biosensor based on Fe₃O₄ NPs–chitosan nanocomposite (Wang, Bao, Yang, Chen, 2008), barium alginate caged Fe₃O₄/C₁₈ magnetic NPs (Zhang, Shi, Yang, De, & Wang, 2010), polyaniline-coated nano-Fe₃O₄/carbon nanotube composite as the protein digestion (Wang, Tan, Zhao, & Liu, 2008), Fe₃O₄/PPy/P(MAA-co-AAm) trilayered composite (Luo et al., 2010), Fe₃O₄–PVA nanocomposites (Liu, Hu, Liu, Liu, & Chen, 2008), polyaniline/nano-Fe₃O₄/composites (Xiao, Tan, Ji, & Xue, 2007), polymer–Fe₃O₄ nanocomposites (Yang, Brown, Kempel, & Kofinas, 2008), PEI modified Fe₃O₄/Au NPs (Sun, Zhu, Zhang, Zhang, & Wang, 2010), alginic acid–Fe₃O₄/nanocomposite (Unal, Toprak, Durmus, Sözeri, & Baykal, 2010). The biocompatibility of magnetic materials can be tuned by variety of the polymer coating.

Among the various types of polymers, alginate has attracted intense attention as an important class of biomaterial. It exhibits unique properties such as; biocompatibility, inexpensiveness, mild encapsulation process at room temperature and a relatively inert hydrogel environment within the matrix. Moreover, one of the

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Synthesis and structural characterization of some titanium butoxides modified with chloroacetic acids

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Abstract Reactions of $\text{Ti}(\text{O}i\text{Bu})_4$ and $\text{Ti}(\text{O}n\text{Bu})_4$ with chloroacetic acids (mono-, di-, and tri-) were carried out in toluene in 1:1 and 1:2 molar ratios at room temperature to generate new precursors of titania. These modified alkoxides/oxo-alkoxides were characterized by spectroscopic methods. A solution of $\text{Ti}(\text{O}n\text{Bu})_2(\text{OCCCCl}_3)_2$ in toluene left for crystallization at -30°C yielded single crystals of the product $\text{Ti}_6(\mu_2\text{-O})_2(\mu_3\text{-O})_2(\mu_2\text{-OC}_4\text{H}_9)_2(\text{OC}_4\text{H}_9)_6(\text{OCCCCl}_3)_8$. Single-crystal X-ray diffraction revealed the molecular structure to be composed of a hexanuclear unit in which two Ti_2O_3 units (made by two edge-sharing octahedra) are linked with two corner-sharing octahedra.

Introduction

Titanium alkoxides have been among the most studied alkoxides during the last decades due to their scientific and technological importance [1–4]. They are widely used as the chief precursors in the sol–gel process for producing TiO_2 and related materials. Reactivity of titanium alkoxides toward water is much higher than that of silicon alkoxides because of the lower electronegativity of titanium in

comparison with silicon and also due to the possibility of higher coordination numbers (c.n.) for titanium. The high hydrolytic instability of metal alkoxides need to be overcome by the incorporation of hydroxylated nucleophilic ligands such as carboxylic acids and β -diketones to produce new molecular precursors [5]. These ligands impart chemical and structural consequences for sol–gel processing. In addition to controlling the reactivity of the metal alkoxides, these ligands also introduce functional organic groups into gels.

The key to “designing” sol–gel materials by the proper choice of the precursors is based on understanding of the chemical reactivity and the structural chemistry of the molecular precursors. Compared with most other metals, the structural chemistry of titanium alkoxides is simpler because the coordination number of titanium in stable compounds is mostly six and sometimes five, but rarely larger than six. The most studied member of this group is titanium isopropoxide, $[\text{Ti}(\text{OPr}^i)_4]$ which is an oil; hence, the actual solid state structure has not been determined [5, 6]. Reportedly, the solution state of most of the titanium alkoxides $[\text{Ti}(\text{OR})_4]$ ($\text{R} = \text{Me}$, Et , and Bu^n) in organic solvents seems to be an equilibrium between mono-, di-, and trinuclear species, promoting the trinuclear species at room temperature with pentacoordinated metal centers [7, 8]. When the steric bulk of the alkoxide is increased ($\text{OR} = \text{OPr}^i$ or $\text{O}i\text{Bu}^i$), generally monomeric species are established, with tetrahedral coordination of Ti [6, 7]. $\text{Ti}(\text{OPr}^i)_4$ has an aggregation of 1.4, indicating that some degree of oligomerization occurs in solution [5]. This variation in the actual solution structures along with a dependence on temperature of solution and solvent makes it difficult to predict the characteristics of $\text{Ti}(\text{OPr}^i)_4$ in a given system. Therefore, well-characterized, highly soluble homoleptic titanium alkoxide complexes are of interest to a wide range of investigators.

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Synthesis, Characterization and Photoluminescent Property of a Trinucleated Cadmium (II) Coordination Polymer Involving In Situ Ligand Reaction

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Abstract A novel two-dimensional cadmium complex $[\text{Cd}_3(\text{pzta})_2(\mu\text{-Cl})_4\text{H}_2\text{O})_2]_n$ (**1**) has been synthesized hydrothermally by the reaction of pyrazine carbonitrile and sodium azide in the presence of cadmium(II) chloride as Lewis acid catalyst. Complex **1** was characterized by elemental analysis, thermal gravimetric analysis, IR spectroscopy and single-crystal X-ray diffraction. Complex **1** crystallizes in triclinic crystal system, having the $P\bar{1}$ space group, which by utilizing hydrogen-bonding extends the molecular complex into 3D hydrogen-bonded metallo-supramolecular networks. It has been shown that N–H–O hydrogen bond based on $R_2^2(26)$ and $R_2^2(12)$ supramolecular synthons can be relied upon as effective tools for the assembly and organization of hexa-coordinated complex. Furthermore, compound **1** shows strong photoluminescence properties at room temperature in solid state.

Keywords Pyrazine tetrazolate (pzta) · In situ · Hydrothermal · Framework

Introduction

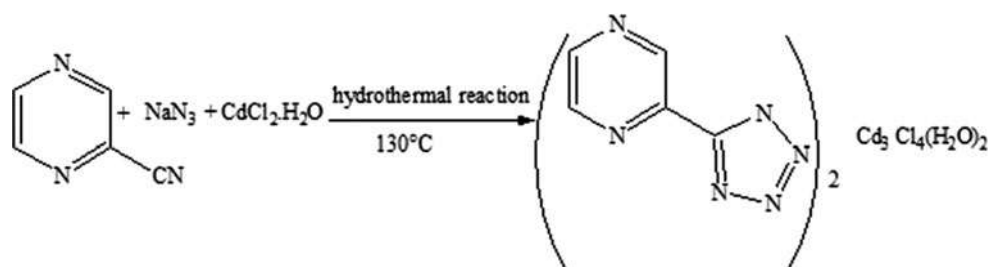
Metal–organic frameworks have attracted much attention owing to their enormous variety of interesting framework topologies and wide application such as gas storage and separation, catalysis and sensors [1–14]. Hydrogen bonding

interactions between moieties are the well-developed methods to increase structural dimensionality of supramolecular systems in the crystal engineering ‘toolbox’ [15–17]. The main purpose of crystal engineering is the understanding of intermolecular interactions and packing principles in molecular crystals with the final goal to use this knowledge for deliberate design of novel materials with tailored structures and properties [18–23]. With this goal in mind, a large number of crystal structures has been evaluated during the last two decades and a series of frequently recurring supramolecular motifs has been identified. Most of these synthons use hydrogen bonds as molecular glue and oxygen, nitrogen, or halides as donors and acceptors, which are embedded into stiff molecular frames [22]. The self assembly of metal–organic frameworks is highly influenced by factors such as solvent system, pH value of solution, coordination nature of the metal ion, structural features of the polydentate organic ligand, metal/ligand ratio and possible second organic ligand [23–25]. Therefore, much more work is required to extend our knowledge of the relevant structural types and establish proper synthetic strategies that lead to the desired species with predictable structures and properties.

In recent years, polydentate aromatic nitrogen heterocyclic ligands with five-membered rings (azoles) have been well used for the construction of supramolecular structures [26–30]. Among polyazoles, imidazoles and triazoles ligands have been extensively used to construct various coordination networks with diverse topologies and interesting properties [30–33]. Tetrazoles exhibit a strong network ability. However, they usually act as mono- or bidentate ligand in most of the reported adducts [34, 35]. The 5-substituted tetrazole possessing acidity close to the carboxylate group, its four nitrogen atoms being able to exhibit versatile bridging modes, has been extensively used in medicinal, coordination and material chemistry.

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Scheme 1 Reaction scheme pertaining to the synthesis of complex **1****Table 1** Crystallographic data and processing parameters for Compound **1**

Empirical formula	C ₁₀ H ₁₀ N ₁₂ Cd ₃ Cl ₄ O ₂
Formula weight	809.33
Temperature	293(2)
Crystal system	Triclinic
Space group	<i>P</i> $\bar{1}$
<i>a</i> /Å, <i>b</i> /Å, <i>c</i> /Å	6.9657(4), 7.5934(5), 11.0040(7)
α /°, β /°, γ /°	101.339(5), 96.824(5), 109.725(6)
Volume/Å ³	526.27(6)
<i>Z</i>	1
ρ_{calc} /mg/mm ³	2.554
μ /mm ^{−1}	3.548
<i>F</i> (000)	382
Crystal size (mm)	0.05 × 0.28 × 0.2
Theta range for data collection	4.22 to 26.31°
Index ranges	−8 ≤ <i>h</i> ≤ 8, −9 ≤ <i>k</i> ≤ 7, −13 ≤ <i>l</i> ≤ 12
Reflections collected	3,409 <i>z</i>
Absorption correction method	Multi-scan
Independent reflections	2,129 [<i>R</i> (int) = 0.0127]
Data/restraints/parameters	2129/0.2796P/152
Goodness-of-fit on <i>F</i> ²	1.052
Final <i>R</i> indexes [<i>I</i> > 2σ(<i>I</i>)]	<i>R</i> ₁ = 0.0162, <i>wR</i> ₂ = 0.0421
Final <i>R</i> indexes [all data]	<i>R</i> ₁ = 0.0174, <i>wR</i> ₂ = 0.0426
Largest diff. peak/hole	0.506/−0.556

Our productive avenue is to use pyrazinyl tetrazolate ligand, as it is good bridging group that can promote the formation of polynuclear products. Recently, Sharpless et al. [36–40] reported the click chemistry for the synthesis of a variety of tetrazoles through [2 + 3] cycloaddition reactions of nitriles and azide in the presence of zinc salt as a Lewis acid. This synthetic approach has been extended by varying metal ions in the construction of coordination frameworks. Significant interest has also been arisen in organic and coordination chemistry for the synthesis of novel coordination polymers with ligands generated in situ [36–40].

As a continuation of our studies on novel metal complexes produced during in situ [2 + 3] cycloaddition

reaction systems, we have carried out the cycloaddition reaction of pyrazine carbonitrile with NaN₃ in the presence of CdCl₂ and H₃PO₄. Herein, we report the synthesis, crystal structure, photoluminescence and thermal behavior of a novel cadmium tetrazole complex [Cd₃(pzta)₂(μ-Cl)₄(H₂O)₂]*n* (**1**).

Experimental

Materials and Methods

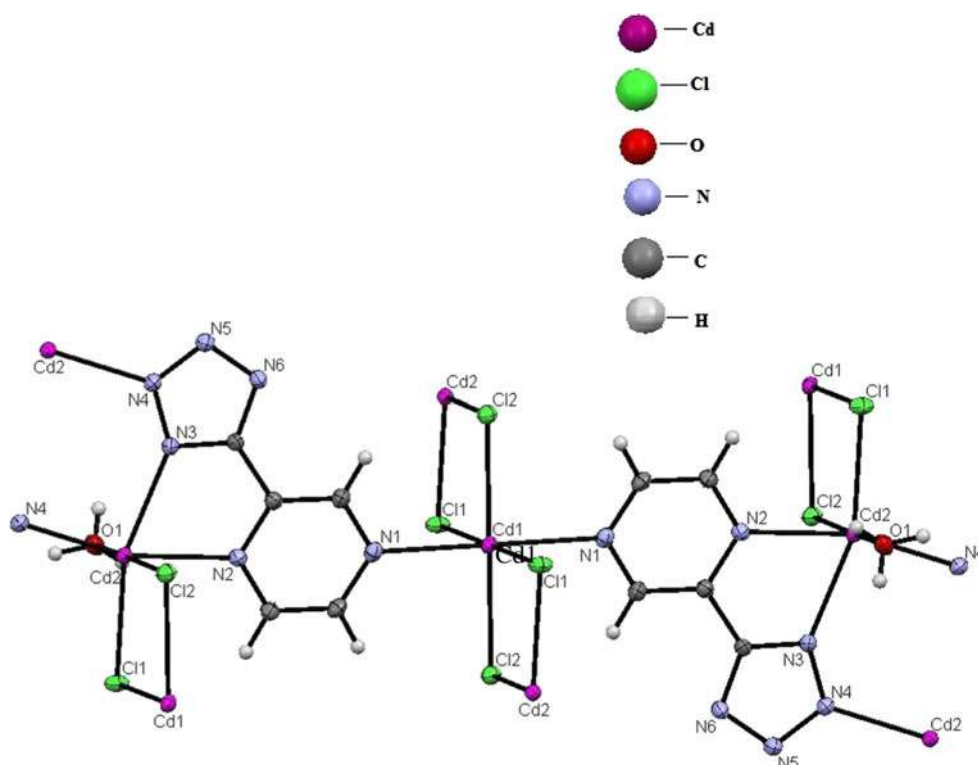
All the starting reagents used in this work were obtained from commercial sources and used without further purification. Thermogravimetric analysis data were collected using a Hi-Res TGA 2950 Thermogravimetric analyzer from TA Instruments. Elemental analyses (C, H and N) were performed on a Perkin–Elmer 2400 CHN Elemental Analyzer. Solution UV–Visible (UV–Vis) spectrum was obtained on a LABOMED spectrophotometer. Infrared spectrum was recorded with a JASCO FT-IR 5300 spectrophotometer (KBr pellet), 4,000–400 cm^{−1} region. The photoluminescence spectrum was recorded on an ocean optics system with range 385–750 nm using excitation wavelength of 385 nm and slit width was 2 nm at room temperature.

Synthesis of [Cd₃(pzta)₂(μ-Cl)₄(H₂O)₂]*n*

The reaction involved in situ ligand transformation of 2-Cyanopyrazine via cycloaddition reaction with the azide ion to give the 5(2-pyrazinyl)tetrazole) ligand.

A mixture of CdCl₂·H₂O (0.201 g; 1.0 mmol) pyrazinecarbonitrile (0.106 g; 2.0 mmol), NaN₃ (0.65 g; 1.0 mmol), H₃PO₄ (0.494 g, 5.0 mmol) and H₂O (15 ml), sealed in a 20 ml Teflon-lined reactor was heated in an oven at 130 °C for 36 h followed by cooling to room temperature at a rate of 10 °C h^{−1} (Scheme 1). The resultant solution was allowed to evaporate slowly at room temperature for several weeks to give colorless needle shaped crystals of **1** (yield: 69 %). Crystals were collected, washed twice with cold water. The obtained crystals were dried in open air for characterization.

Fig. 1 ORTEP diagram showing the coordination atmospheres of Cd atoms in **1**



Crystals were found to be stable under air, light and X-ray exposure.

Microanalysis: calculated for $\text{Cd}_3\text{C}_{10}\text{N}_{10}\text{H}_{12}\text{O}_2\text{Cl}_4$: C, 14.83 %; H, 1.48 %; N, 20.76 %. Found: C, 14.36 %; H, 1.42 %; N, 20.73 %.

Important IR bands (KBr, cm^{-1}): 3430(s), 2048(m), 1654(s), 1529(w), 1460(m), 1411(m), 1284(m), 1145(m), 1072(w), 1037(m), 870(m), 760(m), 640(w), 528(w).

Structure Determination and Refinement

A single crystal suitable for X-ray crystallography was mounted on goniometer head. Complete single-crystal X-ray data of **1** were collected at 293(2) K Fby using a ‘Oxford XCalibur’ diffractometer with Mo $K\alpha$ radiation ($\lambda = 0.71073 \text{ \AA}$). Multi-scan absorption correction was with SADABS [41]. The structure was solved by direct methods and refined through full-matrix least-squares techniques on F^2 by the SHELX-97 program. All non-hydrogen atoms were refined anisotropically, while the hydrogen atoms were located geometrically and refined isotropically. Crystallography data and collection parameters are listed in Table 1.

Crystal data for complex **1**: triclinic, space group $P\bar{1}$ with $a = 6.9657(4) \text{ \AA}$, $b = 7.5934(5) \text{ \AA}$,

$\lambda, c = 11.0040(7) \text{ \AA}$, $\alpha = 101.339(5)^\circ$, $\beta = 96.824(5)^\circ$, $\gamma = 109.725(6)^\circ$, $V = 526.269 \text{ \AA}^3$, $Z = 1$, $D_c = 2.554 \text{ g/cm}^3$, Mo $K\alpha$ radiation, $\lambda = 0.71073 \text{ \AA}$, $T = 293(2) \text{ K}$,

$\mu = 3.548 \text{ mm}^{-1}$, $GOF = 1.052$, $F_{000} = 382$, $R1 = 0.0162$ [$I > 2\sigma(I)$], $wR2(\text{all data}) = 0.0426$.

Results and Discussion

Description of Crystal Structure

Single crystal X-ray analysis reveals that the **1** crystallizes in the triclinic space group $P\bar{1}$ and can be described as ABAB type framework based on $[\text{Cd}_3(\text{Cl}_4)]$ trinuclear clusters and μ_3 -Pzta ligands. All the three cadmium atoms are in slightly distorted six-coordinated octahedron where the angles subtended at the metal centre vary from $70.96(6)$ to $95.71(4)^\circ$. As shown in Fig. 1, the asymmetric unit of **1** is a centrosymmetric dimer with Cd1 locating at the inversion centre, which consists of one and a half crystallographically independent Cd(II) ions, one bridging pyrazine tetrazole ligand; two chlorine atoms and a water molecule. Both the chlorine atoms are bridging while the water molecule is terminal in position. Two possible orientations of H atoms are observed in H_2O due to disorder. Cd1 is coordinated to four bridging chlorine atoms; two pyrazinyl N donors from two individual pzta ligands. And the octahedral Cd2 is defined by three N donors from the two separate pzta ligands; two bridging chlorine atoms and O of water molecule.

The Cd2–O1 distance is $2.3279(16) \text{ \AA}$ while the Cd–N and Cd–Cl bond lengths are in the range of $2.239(2)$ – $2.417(2)$

Table 2 Selected bond lengths (Å) and bond angles (°) for compound **1**

Cd1–N1	2.4111(18)	N1–C4	1.339(3)
Cd1–N1	2.4111(18)	N1–C1	1.340(3)
Cd1–Cl2	2.6036(5)	N2–C2	1.335(3)
Cd1–Cl2	2.6036(5)	N2–C3	1.342(3)
Cd1–Cl1	2.6275(6)	N3–C5	1.330(3)
Cd2–N4	2.2839(17)	N3–N2	1.342(2)
Cd2–O1	2.3279(16)	N4–N5	1.319(2)
Cd2–N3	2.3575(17)	N4–Cd2	2.2839(17)
Cd2–N2	2.4171(18)	N5–N6	1.335(2)
Cd2–Cl1	2.5500(5)	N6–C5	1.336(3)
Cd2–Cl2	2.6344(5)	C1–C2	1.390(3)
Cl1–Cd2	2.5500(5)	C3–C4	1.390(3)
Cl2–Cd1	2.6036(5)	C3–C5	1.463(3)
N1–Cd1–N1	180.0	N2–Cd2–Cl2	92.07(4)
N1–Cd1–Cl2	89.25(5)	Cl1–Cd2–Cl2	87.379(17)
N1–Cd1–Cl2	90.75(5)	Cd2–Cl1–Cd1	93.618(18)
N1–Cd1–Cl2	90.75(5)	Cd1–Cl2–Cd2	92.227(17)
N1–Cd1–Cl2	89.25(5)	C4–N1–C1	117.12(18)
Cl2–Cd1–Cl2	180.0	C4–N1–Cd1	121.10(14)
N1–Cd1–Cl1	84.29(4)	C1–N1–Cd1	121.59(14)
N1–Cd1–Cl1	95.71(4)	C2–N2–C3	117.17(18)
Cl2–Cd1–Cl1	93.581(17)	C2–N2–Cd2	126.41(14)
Cl2–Cd1–Cl1	86.420(17)	C3–N2–Cd2	116.24(13)
N1–Cd1–Cl1	95.71(4)	C5–N3–N4	104.35(16)
N1–Cd1–Cl1	84.29(4)	C5–N3–Cd2	115.39(13)
Cl2–Cd1–Cl1	86.419(17)	N4–N3–Cd2	140.06(13)
Cl2–Cd1–Cl1	93.580(17)	N5–N4–N3	110.08(16)
Cl1–Cd1–Cl1	180.0	N5–N4–Cd2	125.09(13)
N4–Cd2–O1	91.68(6)	N3–N4–Cd2	124.81(13)
N4–Cd2–N3	95.05(6)	N4–N5–N6	108.55(16)
O1–Cd2–N3	91.60(6)	N5–N6–C5	105.41(17)
N4–Cd2–N2	164.31(6)	N1–C1–C2	121.4(2)
O1–Cd2–N2	83.96(6)	N2–C2–C1	121.5(2)
N3–Cd2–N2	70.09(6)	N2–C3–C4	121.32(19)
N4–Cd2–Cl1	104.53(5)	N2–C3–C5	115.41(18)
O1–Cd2–Cl1	90.39(4)	C4–C3–C5	123.25(19)
N3–Cd2–Cl1	160.26(4)	N1–C4–C3	121.4(2)
N2–Cd2–Cl1	90.61(4)	N3–C5–N6	111.60(18)
N4–Cd2–Cl2	92.75(5)	N3–C5–C3	121.83(18)
O1–Cd2–Cl2	175.42(4)	N6–C5–C3	126.50(18)
N3–Cd2–Cl2	89.17(4)		

Symmetry transformations used to generate equivalent atoms

#1 $x, y + 1, z$ #2 $-x, -y + 2, -z$ #3 $-x, -y + 1, -z$ #4 $x, -y + 3/2, z + 1/2$

and 2.5500(7)–2.6344(7) Å respectively, which are comparable to those reported values [42–46]. The bond angles O–Cd2–N around Cd2 are in the range of 83.96–91.68°

while angles O–Cd2–Cl range from 90.39 to 175.42°. In **1** the bond angles N–Cd2–N are in the range from 70.09 to 180° while the angles fall under the limit of 86.42°. Selected bond lengths and bond angles are given in Table 2.

One Cd(1) and two Cd(2) are linked by two bridging Cl atoms to form a [Cd₃Cl₄] trinuclear cluster. Further, the trinuclear clusters are connected by μ_3 -Pzta, which ultimately leads to the formation of the two dimensional coordination polymer in the *b, c*-plane, showing two distinct holes in each sheet, the larger hole of the dimension 11.853 × 9.361 Å and the smaller hole of the dimension 8.578 × 8.395 Å (based on Cd–Cd distance) as shown in Fig. 2. Each pzta acts as a tetradentate bridging ligand, in which the both tetrazolate and pyrazinyl groups adopt μ_2 coordinating mode.

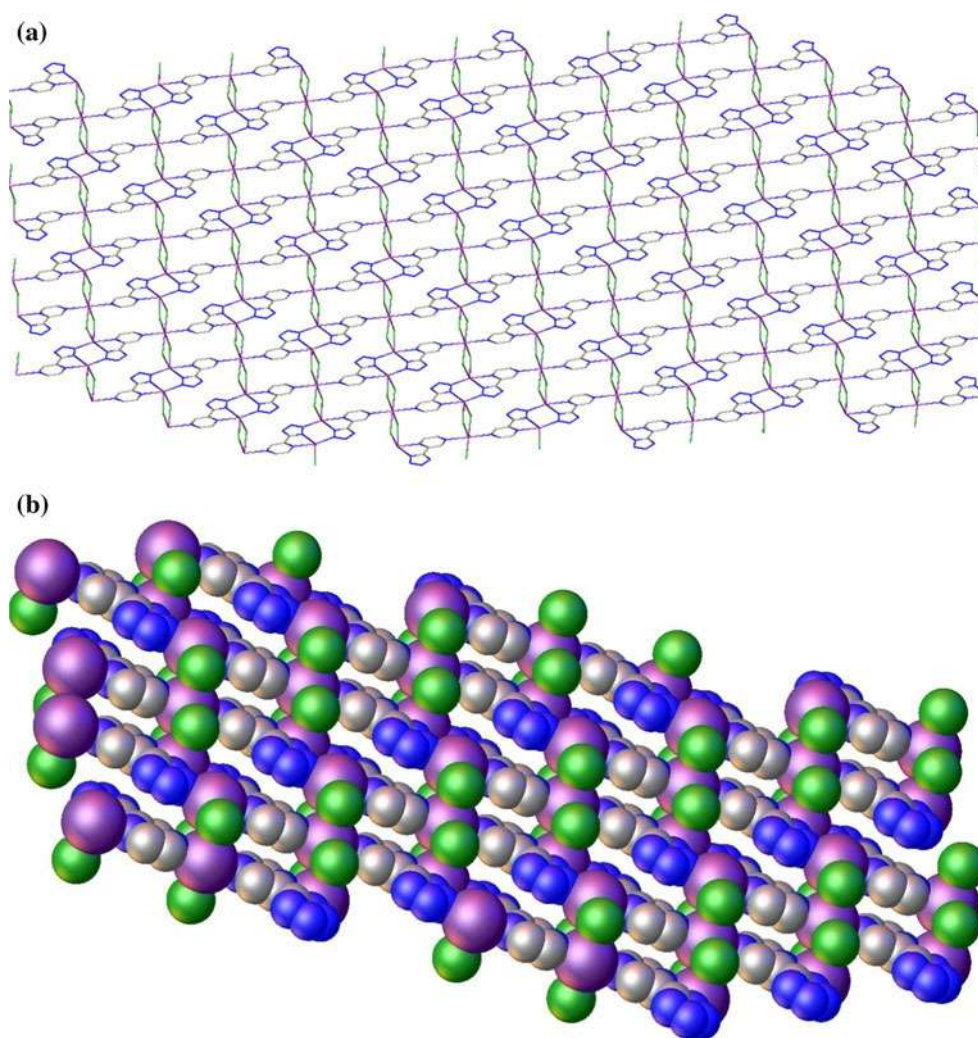
The crystal structure of **1** shows the self organizing behavior of [Cd₃(pzta)₂(μ -Cl)₄H₂O]₂ building blocks and a variety of hydrogen bonding synthons formed via intermolecular neutral hydrogen bonding between the coordinated water molecules and non-coordinated nitrogen atoms of the tetrazole to configure a supramolecular microporous 3D network. The coordinated water molecules act as H-bond donor while tetrazolyl nitrogen atoms behave as acceptors.

The hydrogen bonded site in crystal lattice of complex **1** is oriented along the *b*-axis and linked by wO–H–N bonds (between the bridging water molecule and tetrazole ring) via R_2^2 (26) and R_2^2 (12) supramolecular synthons (Fig. 3). Formation of both types of supramolecular synthons results in the establishment of a 3D supramolecular network along the *b*-axis (Fig. 4). In **1**, the N–O bond distance is 2.727 Å, and the N–H–O bond angle of the intermolecular hydrogen bond is 164.81°.

It is noteworthy that in the crystal packing of **1**, the noncovalent binding forces play an important role in the stabilization of the complex and propagation of the low dimensional structure to a 3D framework due to robust hydrogen bonding interactions involving a variety of supramolecular synthons such as R_2^2 (26) and R_2^2 (12). The 2D network is joined in the third dimension by wO–H–N interactions. Thus, the supramolecular synthons R_2^2 (26) and R_2^2 (12) appear to have reliable utility for constructing a 3D supramolecular structure from 2D coordination chain.

A number of Cd(II) coordination polymers from tetrazolates have been obtained, but both composition and structural features of **1** are different from those reported. For example, complexes [Cd(pzta)₂]*n* and [Cd₂(pzta)(OH)(SO₄)]*n* [47] were formed by the reaction of the CdSO₄·(8/3)H₂O with NaN₃ and pzCN (pz = pyrazine) in different synthetic conditions. The complex [Cd(pzta)₂]*n* is a one-dimensional chain while [Cd₂(pzta)(OH)(SO₄)]*n* is a three dimensional open framework constructed by pzta

Fig. 2 View of the crystal lattice in the *b*, *c*-plane showing: **a** ABAB type between two adjacent layers in **1** (For clarity, some atoms in ligand are omitted). **b** the space filled diagram showing two types of microporous cavities of the dimension $11.853 \times 9.658 \text{ \AA}$ and $8.578 \times 8.395 \text{ \AA}$



ligands associated with sulfate anions. $\text{Cd}(\text{Hpytz})_2(\text{H}_2\text{PO}_4)_2(\text{H}_2\text{O})_2$ ($\text{Hpytz} = 5\text{-(4-pyridyl)tetrazole}$) [48] was prepared under hydrothermal condition and the complex exhibited a 3-D supramolecular array formed by π – π stacking interactions and hydrogen-bonds. In this complex two H_2PO_4 groups are coordinated with the $\text{Cd}(\text{II})$ along with two Hpytz ligands and two water molecules. Structural difference can be attributed to different ligands used and different experimental conditions.

Infrared Spectroscopy

The IR spectrum of the title complex (**1**) exhibits several characteristic strong and medium bands (Fig. 5). The ligand has several potential donor sites which have the tendency to coordinate with the metal ions. IR spectrum of the cadmium complex, $[\text{Cd}_3(\text{pzta})_2(\mu\text{-Cl})_4(\text{H}_2\text{O})_2]_n$, the absence of absorptions around $2,300 \text{ cm}^{-1}$ indicates the absence of cyano while the peaks in the range of $1,557\text{--}1,447 \text{ cm}^{-1}$ indicate that the [2 + 3] cycloaddition

reaction between cyano group and azide anion has taken place. A broad band over $3,100 \text{ cm}^{-1}$ assigned to the presence of coordinated water molecules.

XRPD, TGA and Photoluminescent Property

The simulated and experimental XRPD patterns obtained at room temperature of compound are shown in Fig. 6. Their peak positions are in good consistency with each other indicating the phase purity of the solids. The difference in intensity may be due to preferred orientation of the powder samples.

Compound **1** was analyzed by thermogravimetric analysis (TGA) under inert atmosphere, while ramping the temperature at a rate of $10 \text{ }^\circ\text{C min}^{-1}$ from 0 to $1,100 \text{ }^\circ\text{C}$ (Fig. 7). The thermal decomposition profile of **1** indicated that it was stable up to $280 \text{ }^\circ\text{C}$ followed by a rapid weight loss in the temperature range $280\text{--}336 \text{ }^\circ\text{C}$ corresponding to the loss of two interstitial or coordinated water molecule (calc. 4.44, obsd. 4.18 %). The weight loss between

Fig. 3 H-bonded open metallo-supramolecular network showing supramolecular synthons

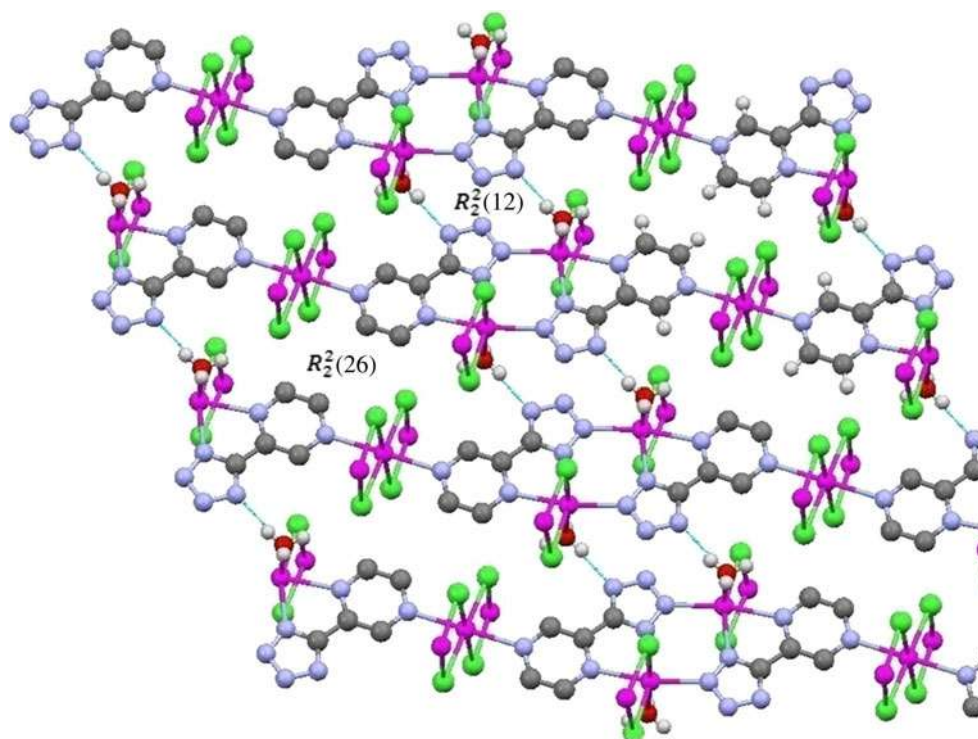
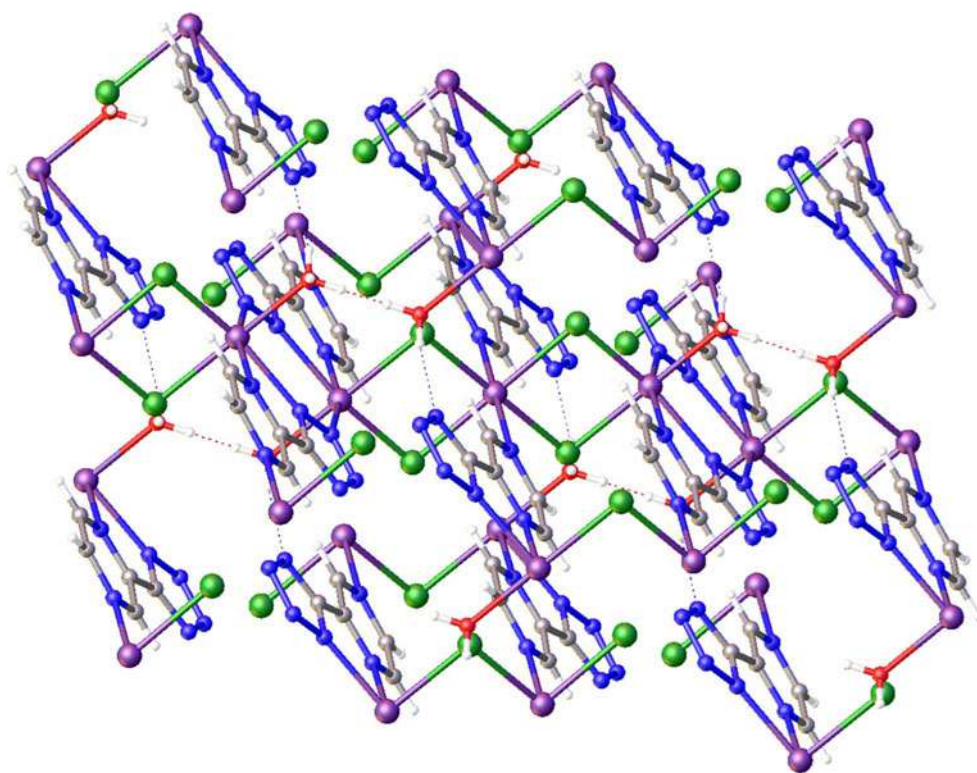


Fig. 4 View of the crystal lattice showing the 3D H-bonded metallo-supramolecular network along the *b*-axis



378–388 °C has been attributed to the decomposition of one tetrazolate group and four Cl atoms. (calc. 26.61, obs. 25.57 %). In the temperature range 432–617 °C, 1.5 molecules of N₂ and one pyrazine group were liberated (calc. 21.52, obs. 20.92 %). Finally remaining pyrazine group was

released from the metal complex and the cyano group left on the metal atom was further oxidized to give the cadmium oxide residue [49].

In DTA graph the exothermic peak at 298 °C signifies the dehydration reaction i.e. loss of water content while at

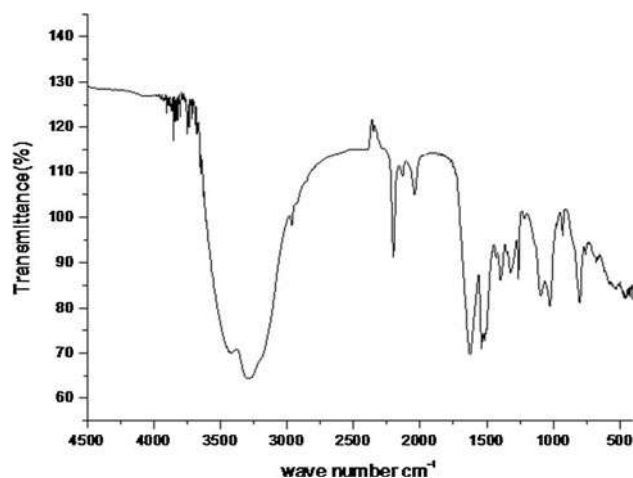


Fig. 5 The IR spectrum of complex 1

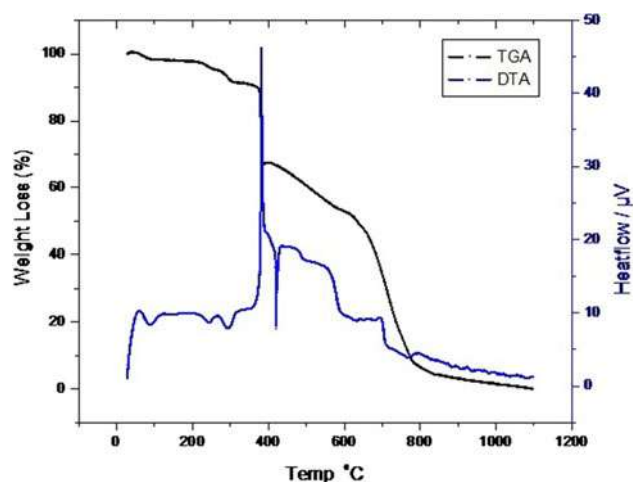


Fig. 7 TG and DTG curves for complex 1

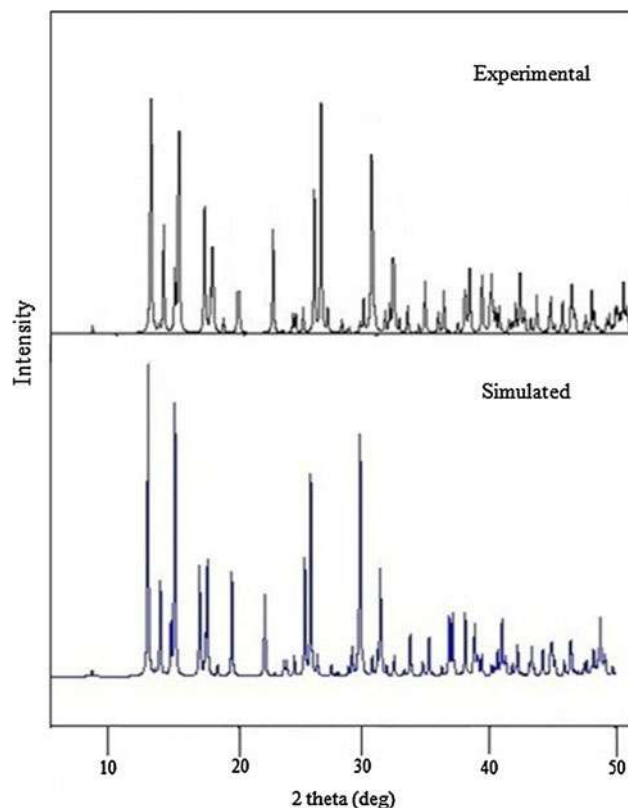


Fig. 6 XRPD patterns for **1**: bottom, calculated from single crystal X-ray data; top, experimental data

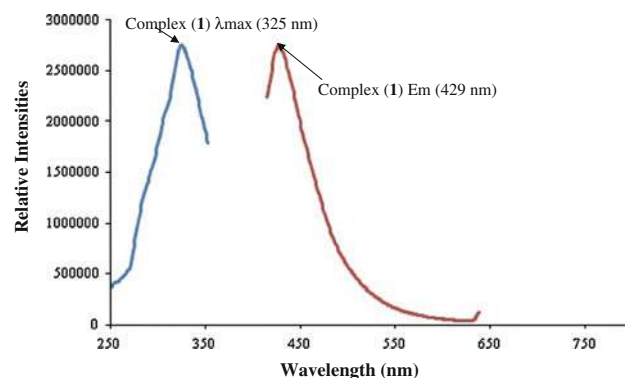


Fig. 8 The absorption and solid-state fluorescent spectra of **1** at room temperature

378 and 420 °C, one tetrazolate group and four Cl atoms were decomposed respectively. The exothermic peaks of the DTA graph show the oxidative nature of the complex **1**.

Among all fluorescent materials with potential applications in light-emitting devices, the d^{10} metal (Cu^I , Ag^I , Au^I , Zn^{II} , Cd^{II}), complexes have been studied most extensively. The pzta ligand with its extended aromaticity

is regarded to be a good candidate for enhanced emissive properties. Photofluorescent measurements of the complex **1** was carried out in the solid state at room temperature. The free pzta ligand exhibits blue fluorescent emission band at 452 nm. As illustrated in Fig. 8, complex **1** upon excitation at 385 nm shows fluorescent emission band with maximum intensity at 429 nm. However, the UV–Vis absorption spectrum of aqueous solution of **1** shows a strong band around 325 nm is tentatively assigned to ligand to metal charge transfer transitions. The highest occupied molecular orbitals (HOMOs) in **1** are presumably associated with the π -bonding orbitals from the aromatic 1,2,3,4-tetrazole rings, whereas the lower unoccupied molecular orbitals (LUMOs) are associated mainly with Cd δ^* -antibonding orbitals, being localized more on the metal centres. Thus, the blue shift in emission peak of **1**, compared with free pzta (452 nm) [50] may be ascribed to the ligand-to-metal charge transfer (LMCT) [51–56]. Since compound **1** has high thermal stability and is virtually

insoluble in most common solvents such as acetone, methanol, chloroform, benzene, water etc., the emission property makes it a potential photoactive material.

Conclusion

In conclusion, we have synthesized and characterized a new Cd-tetraazolate complex of the formula $[\text{Cd}_3(\text{pzta})_2(\mu\text{-Cl})_4(\text{H}_2\text{O})_2]_n$ (**1**), in which the pzta ligand is coordinating in a bridging fashion along with water molecules connecting three metal ions. Interplay of weak non-covalent forces between water molecules and the N atoms of tetrazole results in the formation of a microporous 3D framework. The supramolecular synthons are acting as molecular junctions which play a pivotal role in controlling the crystal packing. This work together with our previous contributions [57, 58], further supplements Sharpless' cycloaddition reactions of azides and nitriles by using Lewis acid as catalyst.

Supplementary Information Available

CCDC 834089 contains the supplementary crystallographic data for **1**. This data can be obtained free of charge via <http://www.ccdc.cam.ac.uk/conts/retrieving.html>, or from the Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: (+44) 1223-336-033; or e-mail: deposit@ccdc.cam.ac.uk.

Acknowledgments This work was supported by the DST project Grant SR/ME/S-3/0016/2008.

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Solution electrophoresis technique in the study of biologically important Fe(III)/ Ni(II) - Norvaline binary complex

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ABSTRACT

A new method solution electrophoresis, involving the use of a ionophoretic technique is described for the study of Fe(III) and Ni(II) biologically important binary complexes with norvaline in solution. The stability constants of ML and ML₂ complexes of Fe(III) – norvaline and Ni(II) – norvaline system have been found to be (9.15 ± 0.05, 7.65 ± 0.04) and (5.48 ± 0.02, 4.55 ± 0.03) (logarithm stability constant values), respectively at 25 °C and ionic strength (μ) = 0.1 M (HClO₄). © 2012 Trade Science Inc. - INDIA

KEYWORDS

Binary complex;
Overall mobility;
Stability constants;
Solution electrophoresis
technique.

INTRODUCTION

The stabilities of binary complexes are known to play an important role in a number of metabolic and toxicological functions. Norvaline is an amino acid (C₅H₁₁NO₂) isomeric with valine, and is commonly made synthetically. It has several significances in biological systems^[1-16]. The importance of metal ions to vital functions in living systems and for the well being of living organisms is now well established. Fe(III) and Ni(II) are well known for their biomedical applications and toxicity^[17-21]. The present modified method is almost free from a number of defects^[22] of common ionophoretic technique such as diffusion, ionic strength and temperature during ionophoresis obviously vitiate the ionophoretic mobility of a particular ion. The technique is very convenient in use.

Recent publications^[23-25] described a new method

for the study of binary complexes by using paper ionophoretic technique. The present work is an extension of the technique, and reports an observation on the determination of the stability constant values of binary complexes of iron(III)/ nickel(II) – norvaline. A simple ionophoretic tube has been designed which after standardization yields remarkable results. Ionophoresis

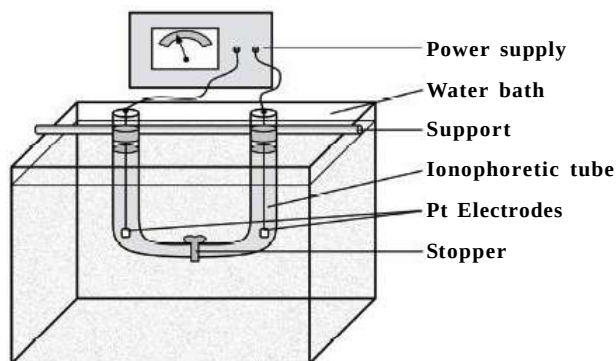


Figure 1: Electrophoresis setup

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set-up is shown in Figure 1.

EXPERIMENTAL

Instruments

Ionophoresis equipment from Systronic (Naroda, India) model 604 was used. It has a built in power supply (a.c.- d.c.) that is fed directly to an ionophoretic tube (18 cm long and 0.5 cm bore) with a stopper in the middle and fused perpendicularly at the ends with short wider tubes of 1.2 cm bore. This tube is kept in a thermostatic water bath at 25 °C. pH measurements were made with century CP 901 digital pH meter using a glass electrode. Absorbance was measured with SPECTROCHEM MK II (PEI) spectrophotometer. A SICO made constant temperature water bath has been used to ensure uniform temperature.

Chemicals

Fe(III) and Ni(II) perchlorate solutions were prepared by precipitating the corresponding carbonates from solution of chlorides (AnalaR grade) with the solution of sodium carbonate, washing the precipitates thoroughly with boiling water and dissolving in a suitable amount of perchloric acid. The resulting solutions were heated to boiling on a water bath and then filtered. The solutions were standardized and diluted with distilled water as required. AnalaR grade ascorbic acid, NaOH, HClO₄ and developing reagent (for Fe(III) 20 % solution of ammonium thiocyanate, for Ni(II) bromine water, concentrated ammonia solution, 1% solution of dimethylglyoxime reagent) for specific colour development were used for different metal ions of binary system sets.

Electrolytic Solutions

For binary systems 15 ml of a solution each containing 1×10^{-3} M Ni(II), 0.1 M HClO₄, and 1×10^{-2} M norvaline was prepared at different pH values (by adding a NaOH solution). In the case of Fe(III), 1×10^{-4} M Fe(III), 0.1 M HClO₄, and 1×10^{-3} M norvaline were used.

Procedure

10 mL of electrolytic solution as mentioned for binary system is taken in an ionophoretic tube and then thermostated at 25 °C. The position of the tube was

adjusted in such a way that the level of the solution in one wide end arm reached a circular mark on it. This adjustment fixed the volume of the solution on both sides of the middle stopper. Two platinum electrodes were dipped in each arm cup and a 50 V potential difference was applied between them. Ionophoresis of the solution was allowed for 45 minutes, after which the middle stopper of the tube is closed. The solution of the anodic compartment was taken out in a 15 mL flask. The Ni(II) content was converted to nickel dimethylglyoximate^[26] and absorbance was taken at 445nm. Similarly the Fe(III) content of the anodic compartment was converted to iron thiocyanate^[26] complex and absorbance was measured at 480 nm against a reagent blank. The observations were repeated for different pH values of different background electrolyte (variation in pH is made by the addition of caustic soda solution). The plot of the absorbance difference vs. pH is shown in Figure 2.

As for the possibility of hydroxy compound formation, we had considered this aspect in our earlier studies. For this we performed the study of binary complex formation at two different concentration of ligand giving complexing species at two different pH values. In that study we find two different plateaus but almost same stability constant. If it would have been a hydroxyl complex only one plateau at fixed pH value must have been formed irrespective of different concentration of ligand (norvaline here) used. This clearly indicates that at very low concentration of metal (i.e. Fe(III) = 10^{-4} M) no hydroxy compound are formed. Hence the possibility of hydroxy compound formation at very low concentrations of metal ions is ignored in these studies.

RESULTS AND DISCUSSION

Chemical literature^[27-30] confirms that anionic species of amino acids are the sole ligating species for metal ions. The electrophoretic mobility of the metal spot against pH gives a curve with number of plateaus is shown in Figure 2. A constant speed over a range of pH is possible only when a particular complex species is overwhelmingly formed. Thus, every plateau is indicative of formation of a certain complex species. The first one in the beginning corresponds to a region in which metal ions are uncomplexed. In this region of low pH, concentration of unprotonated species of norvaline [CH₃

$\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_3^+)\text{COOH}]$ is maximum and this species is non-complexing, beyond this range, metal ions have progressively decreasing mobility, complexation of metal ions should be taking place with anionic species of norvaline, whose concentration increases progressively with increase of pH. Figure 2 shows three plateaus in both Fe(III) and Ni(II), hence both metal ions form two complexes with norvaline anion. Prominent chelating properties have also been assigned to unprotonated anionic species of norvaline, ruling out any such property to the zwitterions^[31-33]. Figure 2 discloses that Fe(III) and Ni(II) metal ions form their first complex movement towards negative electrode.

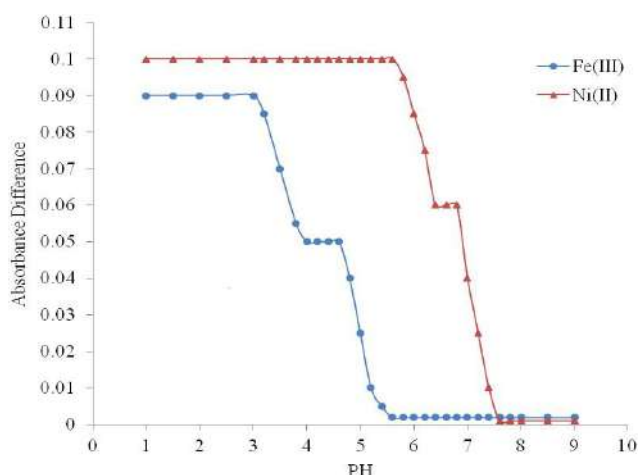
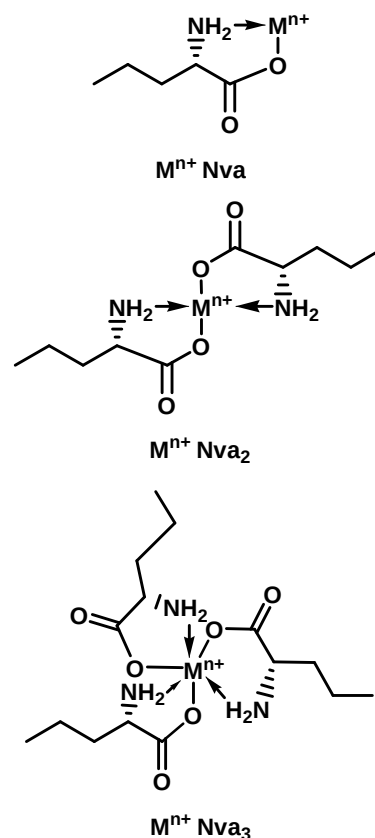


Figure 2 : Mobility curves for the metal-norvaline complexes.

Hence one anionic species of norvaline $[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COO}^-]$, must have combined with Fe(III) and Ni(II) metal ions to give 1:1 $[\text{Fe}\{\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COO}\}]^{2+}$ and $[\text{Ni}\{\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COO}\}]^+$, complex cations, respectively. The third plateau in each case is due to 1:2 metal-ligand complex. Hence, two anionic species of norvaline $[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COO}^-]$, must have combined with Fe(III) and Ni(II) to give 1:2 $[\text{Fe}\{\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COO}\}_2]^+$, and $[\text{Ni}\{\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COO}\}_2]$, metal complexes, respectively. Proposed molecular structures^[35] of binary complexes are given below.

Where $\text{M}^{n+} = \text{Fe}^{3+}$, Ni^{2+} and $\text{Nva} = \text{Norvaline}$ in scheme 1.

Further increase of pH has no effect on the mobility of metal ions, which indicates no further interaction between metal ions and ligands. The complexation of metal ions with norvaline anion may be represented as:



Scheme 1: Molecular structures of binary complexes.



here M^{2+} = metal cations; $[\text{L}^-]$ = norvaline anion; K_1 and K_2 are the first and second stability constants respectively.

The overall mobility U is a composite parameter contributed by different ionic species of the metal ion and is given by following equation (3).

$$U = \frac{u_0 + u_1 K_1 [\text{L}^-] + u_2 K_1 K_2 [\text{L}^-]^2 + \dots}{1 + K_1 [\text{L}^-] + K_1 K_2 [\text{L}^-]^2 + \dots} \quad (3)$$

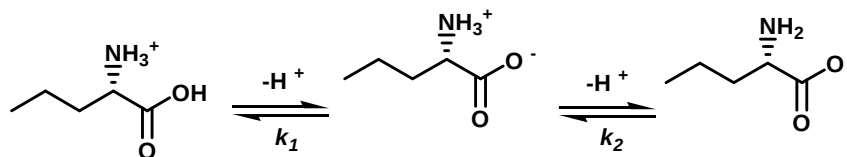
here u_0 , u_1 , u_2 mobilities of the uncomplexed metal ions, 1:1 and 1:2 metal complexes, respectively. The protonation constants of pure norvaline ($\text{pK}_{a1} = 2.31$, $\text{pK}_{a2} = 9.65$)^[34] were determined by same solution ionophoretic technique. The mode of deprotonation of pure norvaline can be represented in scheme 2.

Using protonation constants of pure norvaline the concentration of norvaline anion $[\text{L}^-]$ is determined for the pH value(s) of interest, from which K_1 , can be calculated. The concentration of complexing norvaline anion $[\text{L}^-]$ is calculated with the help of equation.

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$$[L] = \frac{[L_T]}{1 + [H]/k_2 + [H]^2/k_1 \cdot k_2} \quad (4)$$

here $[L_T]$ = total concentration of ligand norvaline $[0.001 \text{ Mol L}^{-1}]$, k_1 and k_2 = first and second protonation constant of pure norvaline, respectively.



Scheme 2 : Deprotonation of pure norvaline

For calculating first stability constant, K_1 , the region between first and second plateau is pertinent. The overall mobility will be equal to the arithmetic mean of the mobility of uncomplex, u_0 , and that of first complex, u_1 , at a pH value where $K_1 = 1 / [CH_3CH_2CH_2CH(NH_2)COO^-]$. The second stability constant K_2 , of 1:2 complex can be calculated by taking into consider-

physiochemical methods reported in chemical literature for the determination of stability constants of metal complexes.

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TABLE 1: Stability constants of binary complexes of Fe(III) and Ni(II) with norvaline.

Metal Ions	Complexes	Stability Constants	Logarithm stability constant values
Fe^{+++}	ML^+	K_1	9.15 ± 0.05
	ML_2	K_2	7.65 ± 0.04
Ni^{++}	ML^+	K_1	5.48 ± 0.02 $5.42^{[34]}$
	ML_2	K_2	4.55 ± 0.03 $4.45^{[34]}$

ation, the region between second and third plateau of mobility curve. The calculated value of K_1 and K_2 are given in TABLE 1.

CONCLUSION

It can be concluded that Fe(III) and Ni(II) are significant for biological systems but as such they are toxic, the norvaline may be used to reduce the level of these metal ions in the biological systems. The present solution electrophoretic technique is very helpful in finding whether complex system is formed or not, if formed its stability constants can also be determined. ML_2 complexes are found to have low stability constant value and less stable in comparison to ML complexes. Stability constants of metal complexes can be very easily calculated through this technique, so the present solution electrophoretic technique has significant advantages over the other

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